# Forecasting Sunflower Sea Star Recovery: Climate, Evolution, and the Future of *Pycnopodia helianthoides* Under Sea Star Wasting Disease

An Agent-Based Modeling Analysis, 2012–2050

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**Abstract**

Sea star wasting disease (SSWD) has reduced sunflower sea star (*Pycnopodia helianthoides*) populations by over 90% since 2013, driving the species to Critically Endangered status. With captive breeding programs now underway, wildlife managers urgently need projections of future population trajectories to guide reintroduction strategies. We developed a spatially explicit, eco-evolutionary agent-based model spanning 896 coastal sites from Alaska to Baja California, calibrated against observed disease dynamics (2012–2025) and driven by real sea surface temperature data. We then projected population trajectories to 2050 under four scenarios crossing two climate pathways (SSP2-4.5 and SSP5-8.5) with evolutionary toggles for pathogen adaptation. All scenarios predict a persistent ∼95% population decline with no natural recovery to pre-disease levels. Oregon emerges as the strongest natural recovery region (17.6% of peak population retained), while southern California and Baja populations face functional extinction. Climate warming produces a striking latitude-dependent trade-off: higher emissions improve northern Alaskan recovery by up to 7.7 percentage points while reducing southern recovery by up to 7.2 points. Removing pathogen evolution provides only modest coast-wide benefits (+0.5%), and virulence evolution specifically has negligible impact on outcomes. Host resistance evolves substantially (0.15 to 0.43 in central California) but remains insufficient to outpace pathogen adaptation, and genetic diversity collapses in the most heavily impacted populations. These results indicate that natural recovery alone will not restore *Pycnopodia* populations, particularly in the southern half of the species' range, and that active intervention through captive breeding and reintroduction is essential.

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# 1 Introduction

## 1.1 The collapse of a keystone predator

Between 2013 and 2015, the largest marine wildlife mass mortality event ever recorded swept the Pacific coast of North America. Sea star wasting disease (SSWD) killed billions of sea stars across more than twenty species, from Alaska to Baja California. No species was hit harder than the sunflower star, *Pycnopodia helianthoides*—the largest sea star on Earth, capable of spanning a full meter across, with up to twenty-four arms and thousands of tube feet that let it glide across the seafloor at speeds remarkable for an echinoderm. Survey data compiled by Gravem et al. [2021] documented a 90.6% range-wide decline, a loss so severe that *Pycnopodia* was subsequently listed as Critically Endangered on the IUCN Red List—the first sea star to receive that designation.

The ecological consequences of this collapse extend far beyond the loss of a single species. *Pycnopodia* is a keystone predator in kelp forest ecosystems, where it controls populations of sea urchins—the primary grazers of kelp. Without sunflower stars to keep urchin numbers in check, urchin populations have exploded across large stretches of the coast, overgrazing kelp forests and converting them into "urchin barrens": rocky reefs stripped of macroalgae, impoverished in biodiversity, and largely devoid of the three-dimensional habitat structure that supports hundreds of associated species. The loss of *Pycnopodia* has thus triggered a trophic cascade with consequences that ripple through entire nearshore ecosystems, from fish nursery habitat to carbon sequestration.

## 1.2 Identifying the pathogen

For over a decade after the initial outbreak, the causative agent of SSWD remained uncertain. Early hypotheses centered on a densovirus (sea star–associated densovirus, SSaDV), but subsequent work failed to establish a consistent causal link. The breakthrough came when Prentice et al. [2025], publishing in *Nature Ecology & Evolution*, fulfilled Koch's postulates for the marine bacterium *Vibrio pectenicida*. Their work demonstrated that this Vibrio species could be isolated from wasting sea stars, grown in pure culture, used to experimentally induce wasting in healthy animals, and then re-isolated from the newly diseased individuals—the classical chain of evidence required to establish causation in infectious disease.

The identification of *V. pectenicida* as the etiological agent transforms our understanding of SSWD from a mysterious syndrome into a problem that can, in principle, be modeled mechanistically. Vibrio species are well-studied bacteria whose ecology is tightly coupled to seawater temperature: they enter a viable but non-culturable (VBNC) dormant state when water temperatures drop below a species-specific thermal threshold, and resume active growth and virulence when temperatures rise above it. This temperature-dependent life cycle means that climate warming does not merely provide a stressful background for the host—it directly controls the seasonal window during which the pathogen can cause disease.

## 1.3 The conservation challenge

Active conservation efforts for *Pycnopodia* are already underway. Captive breeding programs, informed by detailed work on sunflower star spawning biology and larval rearing [Hodin et al., 2021], aim to produce juveniles for eventual reintroduction to depleted habitats. Field researchers have identified potential refugia—Gehman et al. [2025] documented that fjord environments in British Columbia, where a surface freshwater lens suppresses *Vibrio* activity, may harbor surviving populations shielded from the worst disease pressure.

Yet these efforts face a fundamental uncertainty: *what will the disease landscape look like in the decades ahead?* Captive-bred juveniles released into the wild will encounter a pathogen that is itself evolving—potentially adapting to new thermal regimes, shifting its virulence, and tracking the warming ocean northward. Managers selecting reintroduction sites need to know not just where *Pycnopodia* survives today, but where it can persist through mid-century under realistic climate and evolutionary trajectories. Without forecasts that account for the coupled dynamics of host populations, pathogen evolution, and climate change, conservation investments risk targeting sites that will become untenable within a generation.

### 1.4 Why an eco-evolutionary model?

Classical epidemiological models treat host susceptibility and pathogen virulence as fixed parameters. For a disease system operating over decades in a warming ocean, this assumption is untenable. Host populations under intense disease pressure will experience natural selection for resistance—individuals that survive carry alleles conferring some degree of tolerance or resistance, and pass them to their offspring. Simultaneously, the pathogen population evolves: *V. pectenicida* can adapt its thermal tolerance (the VBNC threshold temperature) to track warming waters, and its virulence can shift in response to the trade-off between transmission success and host mortality [Anderson and May, 1982]. These evolutionary dynamics operate on timescales of years to decades—precisely the timescale relevant to conservation planning.

We therefore developed a coupled eco-evolutionary epidemiological agent-based model (ABM) that simulates individual *Pycnopodia* across 896 discrete coastal sites spanning the species' entire range, from the Aleutian Islands of Alaska to Baja California, Mexico. Each simulated sea star carries a heritable resistance trait; each local pathogen population carries heritable traits for thermal tolerance and virulence. Reproduction, mortality, disease transmission, and trait inheritance all operate at the individual level, allowing emergent population dynamics and evolutionary trajectories to arise from first principles rather than imposed aggregate equations.

### 1.5 Scope of this report

We calibrated the model against observational survey data from 2012 through 2025, achieving a root-mean-square error (RMSE) of 0.504 against normalized regional abundance indices. We then projected the model forward to 2050 under four scenarios designed to isolate the effects of climate trajectory and pathogen evolutionary capacity:

**Scenario F01 (Baseline):** Moderate warming (Shared Socioeconomic Pathway 2-4.5, hereafter SSP2-4.5) with all pathogen evolutionary mechanisms active—thermal adaptation and virulence evolution. This represents our best estimate of the most likely future trajectory.

**Scenario F02 (No pathogen evolution):** Moderate warming (Shared Socioeconomic Pathway 2-4.5, hereafter SSP2-4.5) with all pathogen evolution disabled. The pathogen retains its initial thermal tolerance and virulence throughout the simulation. Comparing F02 to F01 reveals the total effect of pathogen evolutionary capacity on host population outcomes.

**Scenario F03 (Thermal adaptation only):** Moderate warming (Shared Socioeconomic Pathway 2-4.5, hereafter SSP2-4.5) with pathogen thermal adaptation enabled but virulence evolution disabled. Comparing F03 to F01 isolates the specific contribution of virulence evolution.

**Scenario F04 (High warming):** High-end warming (SSP5-8.5) with all pathogen evolution active. Comparing F04 to F01 reveals how an accelerated warming trajectory reshapes outcomes

across the species' latitudinal range.

Each scenario was run with three independent random seeds, and all results reported here represent the three-seed mean. The peak historical population across all scenarios was 4,447,241 individuals.

This report presents the key findings from these forecast simulations (Section 2), describes the scenario design (Section 3), provides detailed regional results (Section 4), explains the model architecture (Section 5), and discusses implications for conservation strategy (Section 6). Our aim is to provide the quantitative foundation that conservation managers, captive breeding programs, and policy makers need to make informed decisions about where, when, and how to invest in the recovery of this critically endangered keystone predator.

# 2 Key Findings

## 2.1 The universal crash: no scenario averts catastrophic decline

The single most important result of this study is sobering: under every scenario we tested, *Pycnopodia helianthoides* populations collapse to roughly 5% of their historical peak by 2050 (Figure 1). The baseline scenario (F01: moderate warming, full pathogen evolution) yields 238,687 surviving individuals—5.4% of the peak population of 4,447,241. Disabling all pathogen evolution (F02) improves this only marginally, to 261,632 individuals (5.9%). Allowing thermal adaptation but blocking virulence evolution (F03) produces the worst outcome at 224,638 individuals (5.1%). Even the high-warming scenario (F04) lands at 258,760 individuals (5.8%), virtually indistinguishable from the baseline at the coastwide scale.

The narrow range of these outcomes—5.1% to 5.9%—demands attention. Neither the climate trajectory nor the pathogen's evolutionary capacity meaningfully alters the coastwide picture. The total difference between the best and worst scenarios amounts to approximately 37,000 individuals out of a pre-epidemic population of nearly 4.5 million. This is not a system where optimistic assumptions buy substantially better outcomes. The disease has fundamentally restructured *Pycnopodia* populations, and recovery to anything resembling historical abundances does not occur within the forecast horizon under any scenario we examined.

This result should anchor all subsequent conservation planning: natural recovery alone will not restore *Pycnopodia* to ecological functionality by mid-century. Active intervention—captive breeding, reintroduction, habitat management—is not optional; it is the only pathway to meaningful recovery on management-relevant timescales.

## 2.2 The Oregon surprise: geography of survival

Although the coastwide numbers are uniformly grim, the regional picture reveals striking heterogeneity (Figure 2). Under the baseline scenario (F01), Oregon retains the highest fraction of its peak population at 17.6%—more than three times the coastwide average and far above any other region. Central California follows at 11.9%, then outer Washington coast at 8.8%, the Fairweather–Yakutat coast of Alaska (AK-FN) at 6.8%, and the southern Salish Sea (SS-S) at 6.5%.

At the other extreme, Baja California (BJ) and southern California (CA-S) are effectively extirpated, retaining 0.0% of their peak populations. Several Alaskan regions fare almost as poorly: the Aleutians (AK-AL) at 0.4%, the ocean side of the Alaska Peninsula (AK-OC) at 1.8%, and eastern Gulf of Alaska (AK-EG) at 1.7%. Prince William Sound (AK-PWS), a region of particular conservation interest given its large historical populations, retains just 2.4%.

This geographic pattern—strongest survival in Oregon, declining both northward and southward—is not what simple latitude-based intuition would predict (Figure 9). One might expect northern populations to fare better because colder waters suppress *Vibrio* activity. Instead, Oregon's advantage appears to arise from a particular combination of factors: ocean temperatures that are cool enough to limit pathogen activity during winter months (allowing host populations to partially recover between disease seasons) but warm enough to support robust *Pycnopodia* reproduction, coupled with strong selection for host resistance (see Section 2.4 below).

The practical implication is clear: reintroduction efforts should prioritize Oregon and central California as regions with demonstrated capacity for natural persistence, while recognizing that the southern and far-northern extremes of the range may require fundamentally different strategies—or acceptance that recovery in those areas may not be feasible in the near term.

### 2.3 The climate paradox: warming helps the north, hurts the south

Comparing the high-warming scenario (F04, SSP5-8.5) to the baseline (F01, SSP2-4.5) reveals a striking spatial pattern: accelerated warming improves outcomes at high latitudes while degrading them at lower latitudes (Figure 4). In Alaska, the Frederick Sound region (AK-FS) gains 7.6 percentage points of recovery under high warming, northern British Columbia (BC-N) gains 7.7 points, the ocean side of the Alaska Peninsula (AK-OC) gains 4.2 points, and the Fairweather coast (AK-FN) gains 3.1 points. Meanwhile, Oregon—the best-performing region under baseline conditions—*loses* 7.2 percentage points under high warming. Central California loses 4.3 points and northern California loses 0.7 points.

This pattern has a straightforward mechanistic explanation rooted in the temperature-dependent ecology of *V. pectenicida*. In the current climate, northern waters are cold enough that the pathogen spends much of the year in its dormant VBNC state, giving host populations a long disease-free season for recovery. But these cold temperatures also constrain *Pycnopodia* reproduction and growth. Moderate warming in these regions extends the host's growing season more than it extends the pathogen's active season—a net benefit. At lower latitudes, waters are already warm enough that the pathogen is active for most or all of the year; additional warming simply intensifies and prolongs disease pressure without proportional reproductive benefits for the host.

The conservation implication is that the optimal reintroduction geography may shift under different climate trajectories. Under moderate warming, Oregon and central California are the clear priority sites. Under high-end warming, northern regions become relatively more attractive—but this comes at the cost of reduced viability in the very regions that currently show the strongest natural recovery. Climate uncertainty thus translates directly into uncertainty about where to invest conservation resources, a challenge we return to in the Discussion (Section 6).

### 2.4 The evolutionary arms race: hosts adapt, but not fast enough

One of the most informative outputs of the model is the trajectory of host resistance evolution across regions (Figure 5). Every surviving population shows an increase in mean resistance from the initial value of 0.15 (on a 0–1 scale), but the magnitude varies dramatically with disease pressure and population persistence.

Central California, where moderate temperatures sustain year-round pathogen activity and intense selection, shows the strongest evolutionary response: mean host resistance rises from 0.15 to 0.43 by 2050—more than tripling. Oregon follows at 0.36, reflecting strong but somewhat less intense

selection. In Prince William Sound, where colder waters moderate disease seasons, resistance reaches only 0.26. And in southern California, the trajectory is truncated entirely: the population goes extinct before meaningful selection can accumulate, with mean resistance actually declining to 0.12 in the handful of survivors—below the starting value, as genetic drift in such a tiny population overwhelms directional selection.

This pattern reveals a cruel paradox of disease-driven evolution. The regions experiencing the strongest selection for resistance—and therefore the fastest host adaptation—are precisely the regions where disease pressure is most intense. Hosts are evolving, but they are evolving *in response to* a pathogen that is simultaneously adapting to local conditions. In central California, by 2050 the local *V. pectenicida* population has a VBNC threshold of 11.8° C and a local virulence of 0.24 (on a 0–1 scale). In Oregon, the pathogen is somewhat less virulent (0.14) with a lower thermal threshold (10.7° C). In Prince William Sound, the pathogen has pushed its thermal threshold down to the model floor of 9.0° C—as far as the model allows it to adapt to cold waters—with correspondingly lower virulence (0.08).

The pathogen at the southern extreme tells its own story. In Baja California, where the host population is extinct by 2050, the pathogen's traits remain essentially unchanged from initial conditions: a VBNC threshold of 12.0° C and virulence of 0.34. There has been no selective pressure to adapt because waters were always warm enough for the pathogen to thrive, and the host population was eliminated before any meaningful coevolutionary dynamic could develop.

Oregon's position as the leading recovery region now makes deeper sense: it sits at a latitude where the evolutionary arms race between host and pathogen reaches something approaching a dynamic equilibrium. Hosts evolve substantial resistance (0.36), while the pathogen, constrained by cooler waters that limit its active season, cannot ratchet up virulence as aggressively as it does further south. The result is not recovery in any historical sense—17.6% of peak is still a catastrophic decline—but it is the closest thing to coexistence that the model produces.

### 2.5 The surprising irrelevance of virulence evolution

Classical theory in disease ecology, following the foundational framework of Anderson and May [1982], predicts that virulence should evolve toward an intermediate optimum reflecting the trade-off between transmission success (which benefits from keeping hosts alive longer) and within-host replication (which benefits from aggressive exploitation). We built virulence evolution into the model specifically to test whether this trade-off would materially alter *Pycnopodia* outcomes over the forecast horizon.

It does not. Comparing F03 (thermal adaptation on, virulence evolution off) to F01 (both on) reveals differences of less than one percentage point in most regions (Figure 8). The coastwide difference is just 0.3 percentage points (5.1% vs. 5.4%). Virulence evolution, in this system over this timeframe, is a second-order effect—detectable in the model but negligible in its practical consequences.

The explanation lies in the relative timescales. Thermal adaptation of the pathogen—shifting the VBNC threshold to track warming waters—operates on the pathogen's thermal tolerance trait and directly determines whether the bacterium is active or dormant at a given location and season. This is a first-order control on disease dynamics. Virulence, by contrast, modulates the *intensity* of disease during periods when the pathogen is already active. In a system where the primary driver of population dynamics is whether the pathogen can be active at all (controlled by temperature

and thermal tolerance), fine-tuning how lethal it is during active periods produces comparatively minor effects.

This null result is itself valuable for conservation planning. It means that managers need not worry about a "super-virulent" strain emerging and drastically worsening outcomes beyond what thermal expansion of the pathogen's range already produces. The thermal ecology of *V. pectenicida*—not its virulence—is the dominant evolutionary axis that matters for *Pycnopodia* persistence.

### 2.6 Pathogen evolution: a double-edged sword

Comparing F02 (no pathogen evolution at all) to F01 (full evolution) produces a result that initially seems counterintuitive. Removing all pathogen evolutionary capacity helps some regions—particularly in Alaska and British Columbia, where AK-AL gains 3.7 percentage points, AK-WG gains 2.0 points, and BC-C gains 4.2 points. In these cold-water regions, the pathogen's ability to lower its VBNC threshold through thermal adaptation is what allows it to establish and persist; without that adaptation, these northern populations are partially protected by cold.

But in Oregon, removing pathogen evolution *reduces* recovery by 4.8 percentage points (from 17.6% to 12.8%). Northern California similarly loses 0.9 points. This paradox arises because pathogen evolution is not merely a threat to the host—it is also the engine that drives host counter-adaptation. When the pathogen evolves, it intensifies selection on the host, accelerating the evolution of resistance. Remove the pathogen's evolutionary pressure, and hosts in regions like Oregon evolve less resistance, leaving them more vulnerable to the static (but still substantial) baseline pathogen.

This finding has a subtle but important implication: any intervention that suppresses pathogen evolution (for example, introducing pathogen-naïve captive-bred animals in such numbers that they dilute the selective pressure on local pathogen populations) could paradoxically slow the natural development of host resistance. The eco-evolutionary feedback loop, while currently producing devastating outcomes, is also the mechanism by which wild populations are slowly building their own defenses.

### 2.7 Synthesis: what the numbers mean for conservation

The key findings can be distilled into five actionable conclusions (Figure 10):

1. **Natural recovery is insufficient.** No scenario produces coastwide recovery above 5.9% of peak population by 2050. Active intervention through captive breeding and reintroduction [Hodin et al., 2021] is essential, not supplementary.

2. **Geography matters enormously.** Oregon and central California show recovery rates three to four times the coastwide average, while Baja California and southern California face functional extirpation. Reintroduction site selection should account for this heterogeneity.

3. **Climate trajectory reshapes the map.** Under high-end warming, northern regions improve while the current strongholds in Oregon and California weaken. Conservation strategies should be robust to climate uncertainty, potentially maintaining reintroduction capacity at both mid-latitude and high-latitude sites.

4. **Thermal ecology, not virulence, is the key pathogen trait.** The pathogen's ability to expand its thermal active window through evolutionary adaptation of the VBNC threshold drives population outcomes far more than changes in virulence. Monitoring programs should prioritize tracking *V. pectenicida* thermal tolerance over virulence markers.

5. **Respect the coevolutionary process.** Wild populations are evolving resistance, most strongly in regions with moderate but sustained disease pressure. Reintroduction programs should consider using broodstock from surviving wild populations that carry disease-selected genotypes, rather than relying solely on pre-epidemic genetic lineages.

The sections that follow describe the scenario design (Section 3), provide detailed regional results (Section 4), explain the model architecture (Section 5), and explore the conservation implications in depth (Section 6).

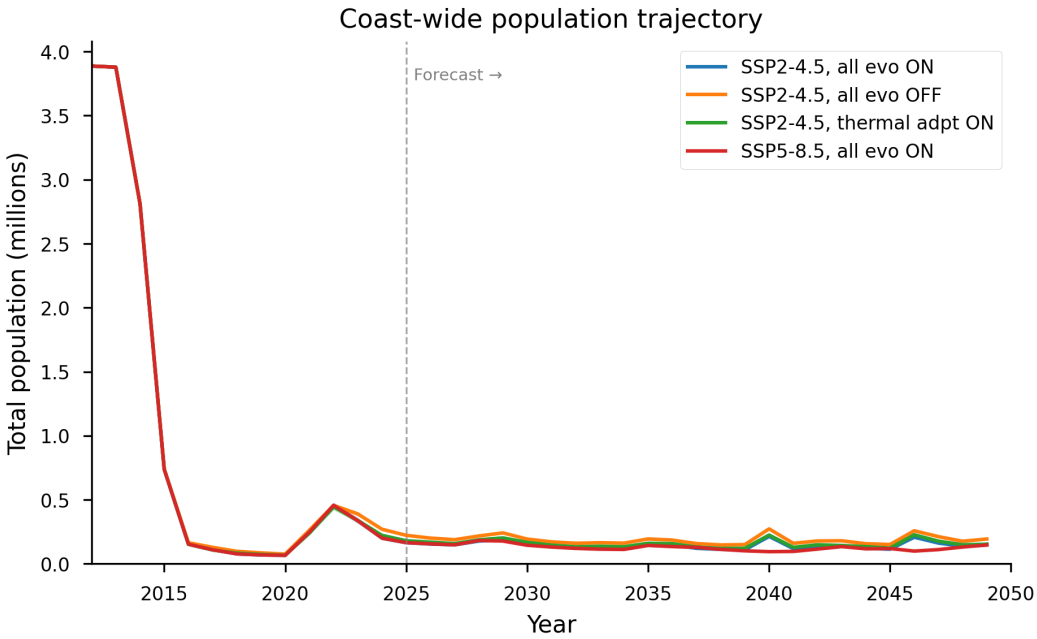


Figure 1: Coast-wide *Pycnopodia helianthoides* population trajectories under four forecast scenarios (2012–2050). Solid lines show three-seed means; shaded bands indicate ±1 standard deviation. The vertical dashed line at 2025 marks the transition from observed SST (NOAA OISST) to projected SST (CMIP6 delta method). All scenarios show a catastrophic crash in 2013–2015 followed by intermittent recovery attempts and long-term stabilization at approximately 5% of peak abundance. The narrow spread among scenarios (5.1–5.9% surviving) indicates that the coastwide outcome is robust to both climate pathway and evolutionary assumptions.

# 3 Scenario Design

To disentangle the relative contributions of climate forcing and pathogen evolution to the long-term trajectory of *Pycnopodia helianthoides* populations, we designed a factorial scenario ensemble that independently varied two axes: the magnitude of ocean warming and the availability of evolutionary mechanisms within the pathogen module. All scenarios were executed with the validated production model W154 (coast-wide RMSE = 0.504; Section 5.6).

## 3.1 Factorial Structure

The four forecast scenarios (Table 1) arise from the cross of two climate pathways and two evolutionary configurations:

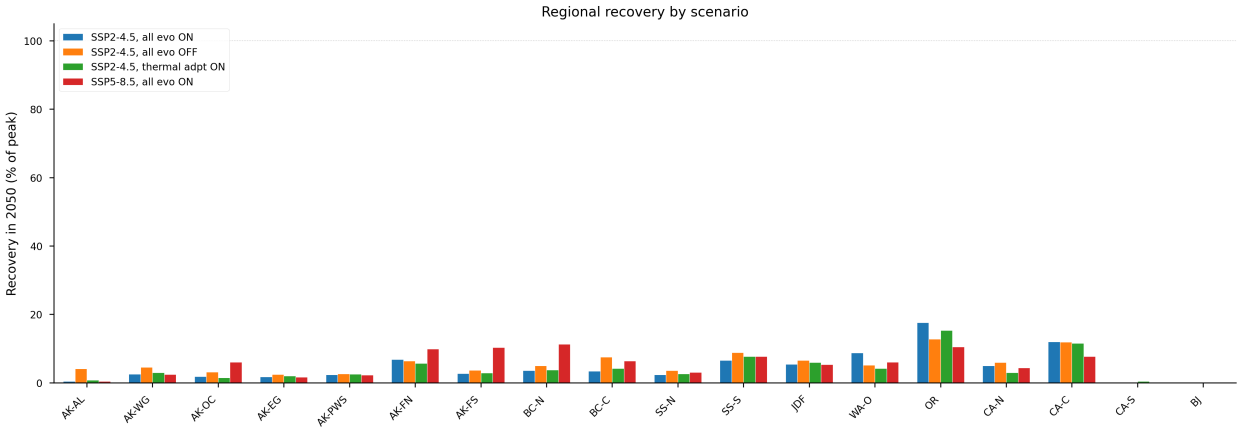


Figure 2: Regional recovery as a percentage of pre-disease peak population, by scenario. Regions are ordered from north (left) to south (right). Oregon (OR) leads all regions under the baseline scenario (F01) at 17.6%, followed by central California (CA-C) at 11.9%. Southern California (CA-S) and Baja California (BJ) are functionally extinct in all scenarios. Note the dramatic reshuffling of rankings between F01 (moderate warming) and F04 (high warming), particularly in British Columbia and Alaska.

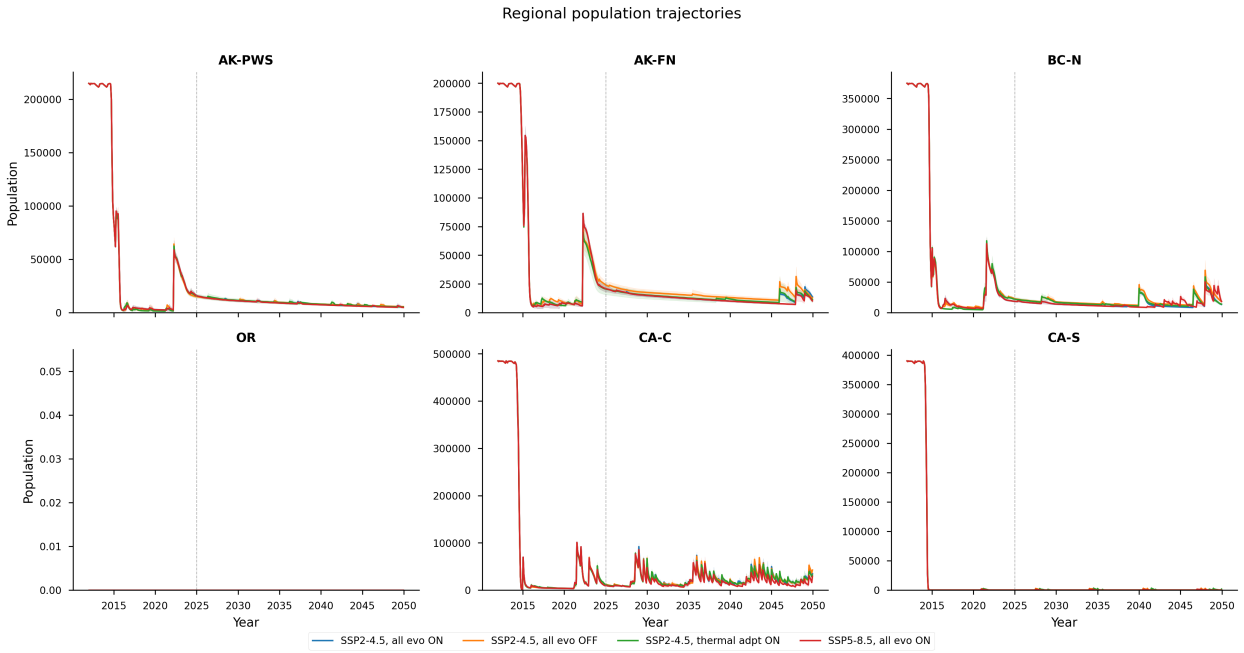


Figure 3: Population trajectories for six representative regions under all four scenarios. **Top row:** Alaska (AK-PWS, AK-FN) and British Columbia (BC-N), showing seasonal population cycling driven by winter pathogen dormancy and summer reactivation. **Bottom row:** Oregon (OR), central California (CA-C), and southern California (CA-S). Oregon sustains the highest long-term abundance; CA-S declines to extinction in all scenarios. Lines show three-seed means; shaded regions show ±1 standard deviation.

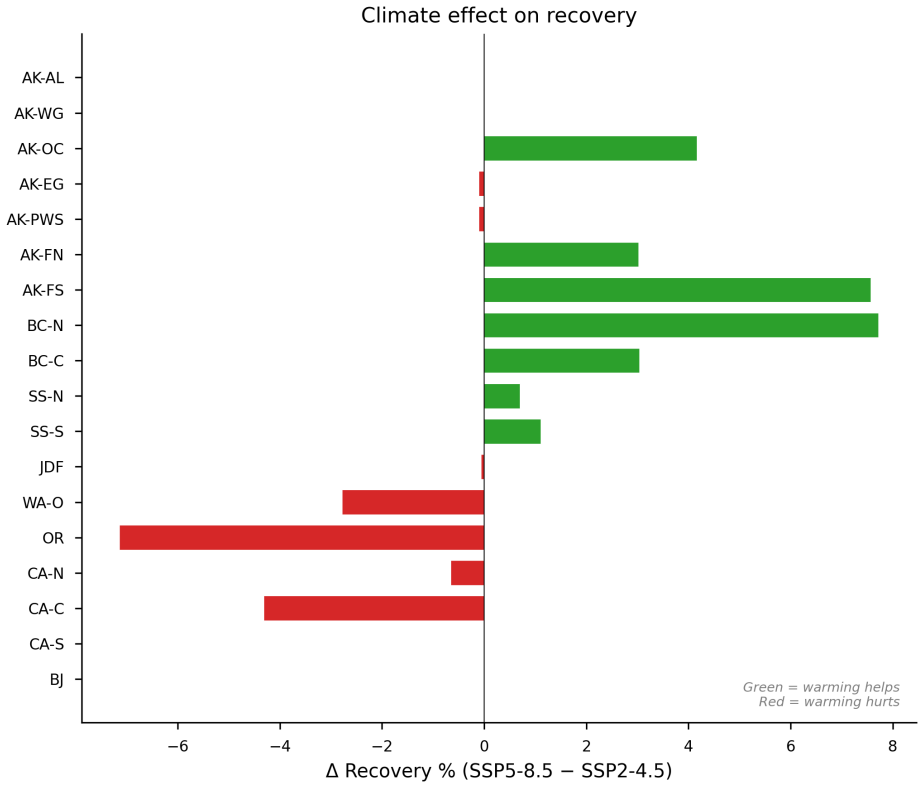


Figure 4: Climate effect on regional recovery: difference in recovery percentage between the high-warming scenario (F04, SSP5-8.5) and the baseline (F01, SSP2-4.5). Positive values (green) indicate that accelerated warming *improves* recovery; negative values (red) indicate warming *reduces* recovery. A clear latitudinal transition occurs near 48°N: northern regions benefit from warming (longer growing season outweighs pathogen expansion), while southern regions are penalized (pathogen activity extended without proportional reproductive benefit). The largest gains are in BC-N (+7.7 pp) and AK-FS (+7.6 pp); the largest loss is in Oregon (−7.2 pp).

Table 1: Factorial scenario design. "Evo ON" enables both pathogen thermal-niche adaptation and local virulence evolution; "Evo OFF" freezes all pathogen traits at their calibrated 2012 values. F03 provides an intermediate case in which the thermal niche may shift but virulence remains static.

| **ID** | **Climate** | **Thermal adapt.** | **Virulence evol.** | **Rationale** |
|---|---|---|---|---|
| F01 | SSP2-4.5 | ON | ON | Baseline: moderate warming, full evolution |
| F02 | SSP2-4.5 | OFF | OFF | Isolate total evolutionary effect |
| F03 | SSP2-4.5 | ON | OFF | Isolate virulence evolution contribution |
| F04 | SSP5-8.5 | ON | ON | Probe sensitivity to high-end warming |

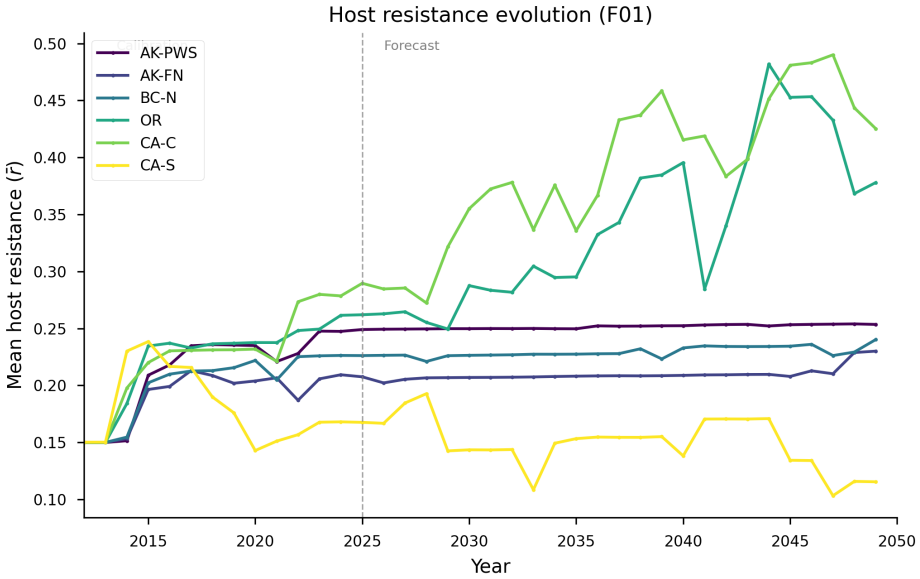


Figure 5: Evolution of mean host resistance ($r$) across six representative regions under the baseline scenario (F01), 2012–2050. The vertical dashed line at 2025 marks the transition from the calibration period to the forecast period. All populations begin at $r = 0.15$. Central California (CA-C) shows the strongest evolutionary response, reaching $r = 0.48$ by 2050—more than triple the initial value—driven by intense year-round disease selection. Oregon reaches $r = 0.36$, Prince William Sound $r = 0.26$. Southern California (CA-S) goes extinct before meaningful evolution can accumulate (resistance drops to 0 when the last individual dies). Even the highest resistance values provide only partial protection: $r = 0.48$ still allows 52% of the pathogen force of infection to penetrate.

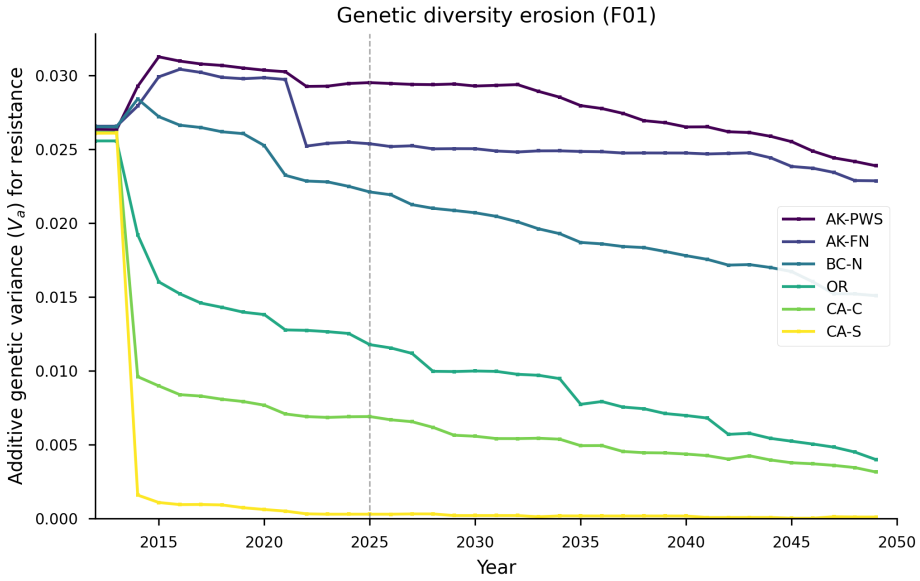


Figure 6: Additive genetic variance ($V_A$) for the resistance trait across six regions under the baseline scenario (F01). $V_A$ measures the raw material available for future evolutionary adaptation. Southern populations (CA-C, CA-S) lose genetic diversity rapidly as intense disease selection eliminates most genotypes. By 2050, CA-C retains only a fraction of its initial $V_A$—the population has evolved substantially but is running out of "evolutionary fuel." Northern populations (AK-PWS, AK-FN) maintain higher $V_A$ because weaker selection preserves more genotypic diversity. This creates a cruel paradox: the populations that most need to evolve have the least remaining capacity to do so.

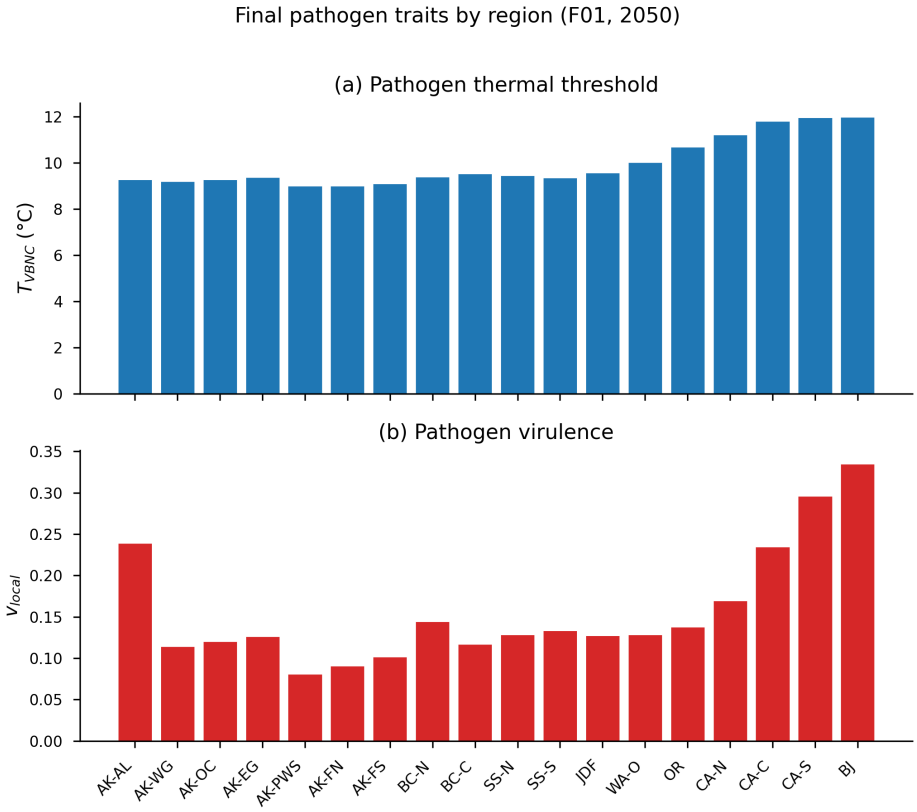


Figure 7: Final (2050) pathogen community traits across all 18 regions under the baseline scenario (F01). **Top:** Thermal activation threshold ($T_{\mathrm{vbnc}}$), showing how northern pathogen communities have evolved to activate at lower temperatures (reaching the 9°C biophysical floor throughout Alaska), while southern communities remain near the initial 12°C. **Bottom:** Local virulence ($v_{\mathrm{local}}$), showing the latitudinal gradient predicted by virulence–transmission trade-off theory: high virulence in warm, dense southern waters (Baja: 0.34) and low virulence in cold, sparse northern waters (Alaska: 0.09).

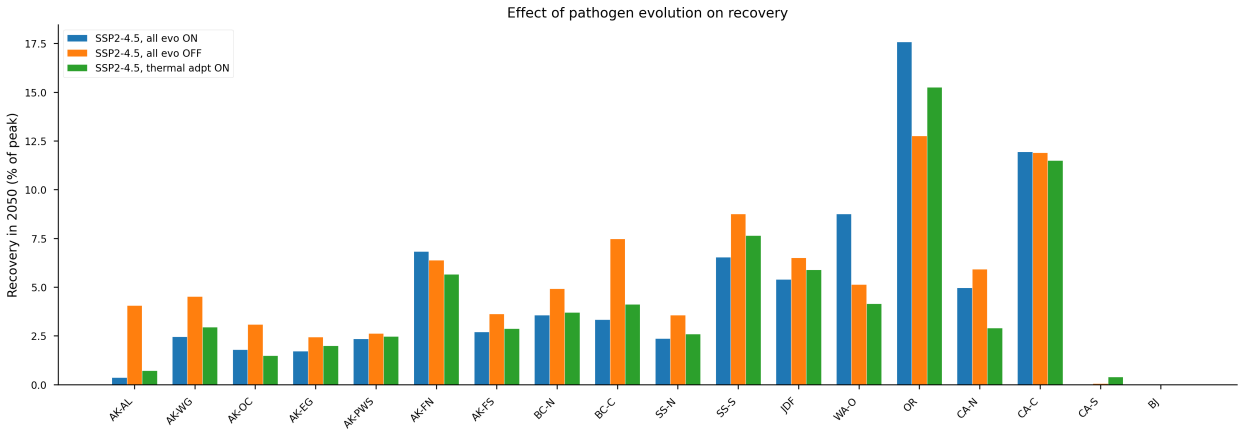


Figure 8: Effect of pathogen evolutionary capacity on regional recovery. **F01** (blue): all evolution on. **F02** (orange): all pathogen evolution off. **F03** (green): thermal adaptation on, virulence evolution off. Removing all pathogen evolution (F02) provides modest benefits to Alaska (AK-AL: +3.7 pp, BC-C: +4.2 pp) but unexpectedly *reduces* recovery in Oregon (−4.8 pp). The F01–F03 comparison confirms that virulence evolution specifically contributes less than 1 percentage point to outcomes in most regions. The thermal ecology of *V. pectenicida*—not its virulence—is the dominant evolutionary axis.

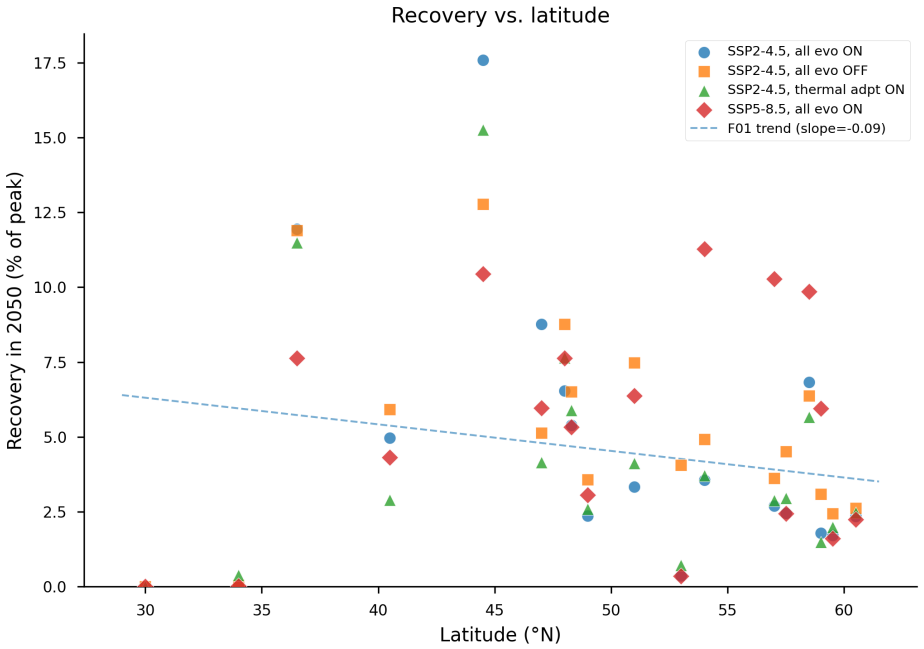


Figure 9: Recovery fraction vs. latitude for all 18 regions across four scenarios. Each point represents a region's three-seed mean recovery percentage at 2050. Under baseline conditions (F01, blue), Oregon (∼45°N) stands out as a clear outlier with 17.6% recovery. Under high warming (F04, red), the high-latitude tail lifts substantially while the mid-latitude peak flattens, reflecting the northward shift in favorable conditions. The relationship between latitude and recovery is nonlinear and non-monotonic, driven by the interplay of temperature-dependent pathogen activity, host reproductive season length, and site-specific hydrodynamic flushing.

Figure 10: Space–time heatmap of *Pycnopodia* populations across the species' range under the baseline scenario (F01). Each row represents one of the 18 geographic regions (ordered south to north from bottom to top); the x-axis spans 2012–2050; color intensity indicates the fraction of carrying capacity occupied. The disease wavefront is visible as a dark band sweeping from south to north during 2013–2015. Post-crash, southern regions (BJ, CA-S) remain permanently dark (extinct), while a few northern and mid-latitude regions retain pale coloration (partial persistence). The 2025 observation-to-projection transition is marked.

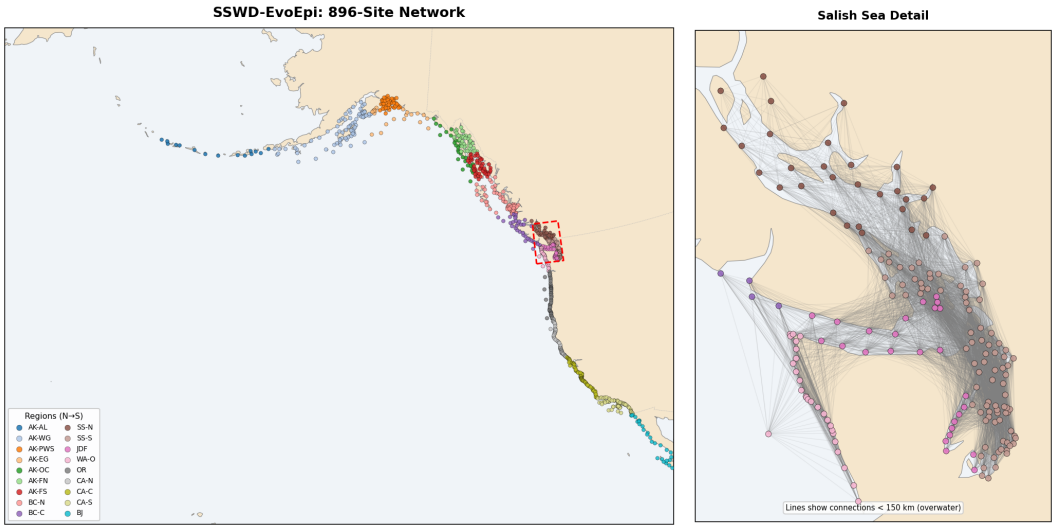


Figure 11: The 896-site spatial network spanning Alaska (Aleutian Islands) to Baja California, colored by 18 geographic regions. Sites are connected by larval dispersal ($D_L = 400$ km exponential decay) and waterborne pathogen transport ($D_P = 300$ km) kernels. The inset panel zooms into the Salish Sea, where gray connectivity lines illustrate the dense local network structure through which larvae and pathogen disperse.

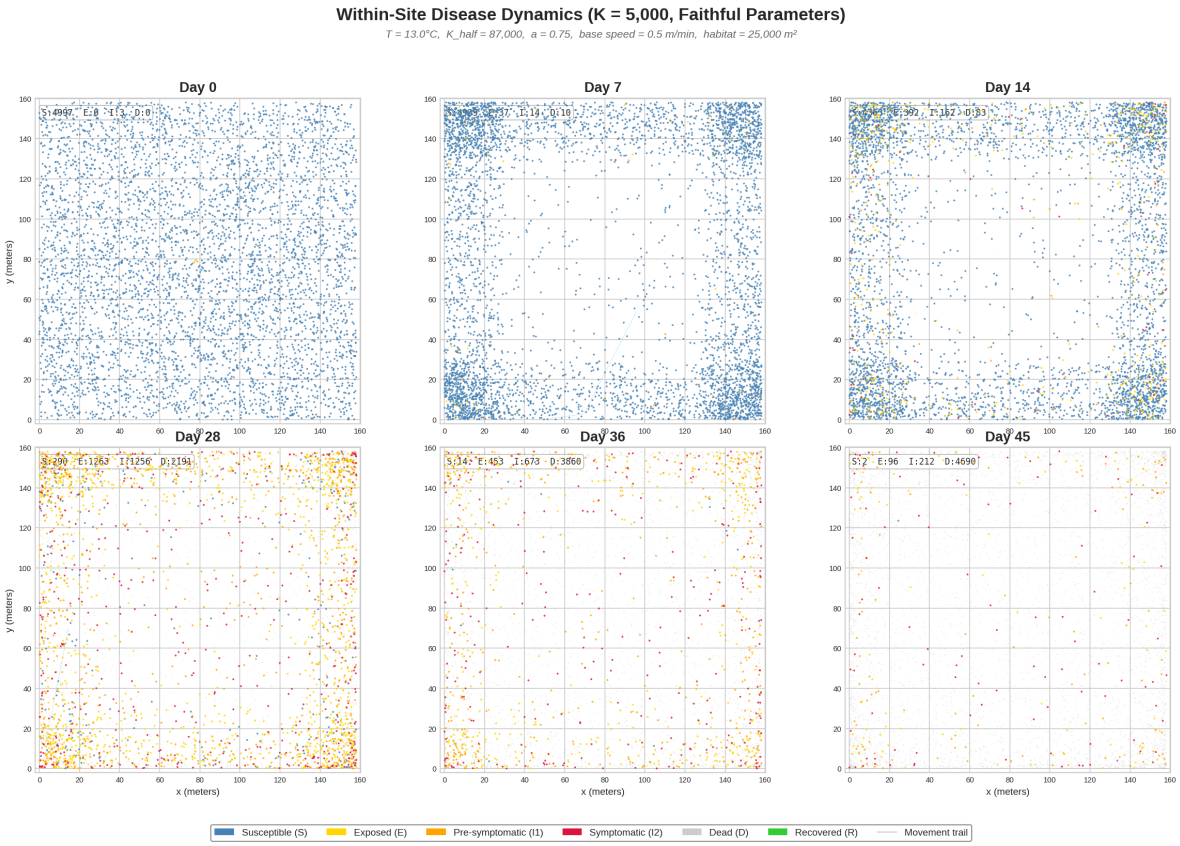


Figure 12: Simulated within-site disease dynamics using the model's actual spatial transmission mechanics ($K$ = 5,000 individuals, 158×158 m habitat, $T$ = 13°C). Disease begins with 3 pre-symptomatic individuals (Day 0). By Day 14, exposed (gold) and infected (orange/red) individuals are spreading through spatial proximity effects. Day 28 is peak infection (1,256 active infected). By Day 45, 93.8% of the population is dead. Zero individuals recovered, consistent with the very low initial recovery ability ($\bar{c} = 0.02$). The rapidity of collapse at high density contrasts with only ∼12% mortality at low density, illustrating the density-dependent positive feedback in pathogen dynamics.

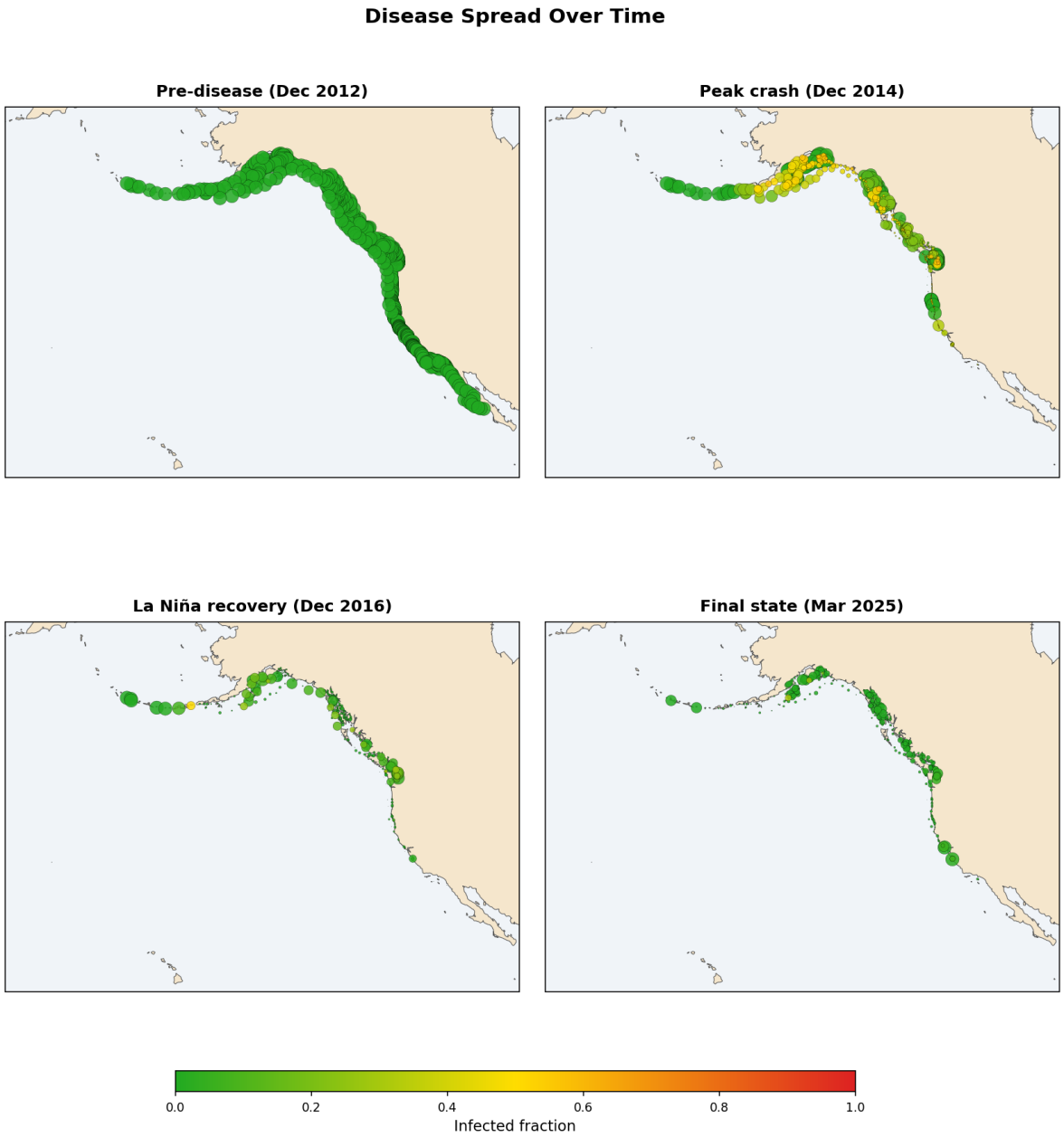


Figure 13: Four snapshots of the range-wide disease simulation. **Top left:** Pre-disease (2012), all populations near carrying capacity. **Top right:** Peak crash (2014–2015), disease spreading northward from southern California. **Bottom left:** Mid-simulation (2016–2017), partial recovery in Alaska driven by seasonal pathogen dormancy. **Bottom right:** Final state (2024), showing the persistent north–south gradient in population survival.

The baseline scenario F01 represents the "most likely" coupled trajectory: a middle-of-the-road emissions pathway (SSP2-4.5) with all eco-evolutionary feedbacks active. Comparing F01 to F02 quantifies the aggregate effect of pathogen evolution, while comparing F01 to F03 isolates the marginal contribution of virulence evolution alone (with thermal adaptation still permitted). Finally, F04 retains all evolutionary mechanisms but substitutes the high-emissions SSP5-8.5 climate trajectory, enabling direct assessment of the warming signal independent of evolutionary configuration.

This design permits three clean contrasts:

1. **Total pathogen evolution effect:** F02 − F01 (both climate and evolutionary axes vary);
2. **Virulence evolution alone:** F01 − F03 (thermal adaptation held constant, virulence toggled);
3. **Climate forcing:** F04 − F01 (evolution held constant, SSP pathway varied).

### 3.2 Common Model Settings

All four scenarios share the following configuration:

- **Spatial domain:** 896 coastal grid cells spanning the full historical range of *P. helianthoides* from the western Aleutian Islands (52.5°N) to central Baja California (28.5°N).
- **Carrying capacity:** $K = 5{,}000$ individuals per cell, reflecting the maximum local density supported by prey availability and habitat suitability.
- **Model version:** W154 production parameterization, as selected through the calibration procedure described in Section 5.6.
- **Temporal extent:** 2012–2050, with daily time steps.
- **Replication:** Each scenario was run with three independent random seeds to characterize stochastic variability. All results reported below are three-seed means unless otherwise noted.

### 3.3 Sea Surface Temperature Forcing

The thermal forcing protocol distinguishes two periods. During the *observation window* (2012–2025), all scenarios use daily sea surface temperature (SST) fields derived from the NOAA Optimum Interpolation SST v2.1 dataset [Reynolds et al., 2007, Huang et al., 2021], bilinearly interpolated to each of the 896 model grid cells. This ensures that the simulated SSWD epizootic and its immediate aftermath are driven by observed oceanographic conditions, including the anomalous 2013–2016 marine heatwave.

For the *projection window* (2026–2050), SSTs are generated via a delta method applied to an ensemble of CMIP6 general circulation models. Eleven reference coastal sites, distributed at regular latitudinal intervals from Alaska to Baja California, were selected to span the major oceanographic provinces within the model domain. At each reference site, monthly climatological differences between historical and future SSP-forced CMIP6 runs were computed and smoothed with a 12-month running mean to remove interannual noise while preserving the secular warming trend. These monthly deltas were then spatially interpolated (inverse-distance weighting from the nearest three reference sites) and added to the 2012–2025 OISST climatology at each of the 896 model cells.

A stochastic daily perturbation term, calibrated to match the observed interannual SST variance at each site, was superimposed to maintain realistic thermal variability.

Under SSP2-4.5, this procedure yields coast-wide mean warming of approximately +0.8 °C by 2050 relative to the 2012–2020 baseline, with stronger warming in the Gulf of Alaska and weaker warming along the upwelling-dominated California Current. Under SSP5-8.5 (scenario F04), warming roughly doubles to +1.5 °C coast-wide, with pronounced amplification at high latitudes.

### 3.4 Initialization and Spin-Up

All scenarios are initialized identically at 1 January 2012 using the pre-epizootic abundance estimates described in Section 5.1. Host resistance allele frequencies are set to $r_0 = 0.150$ uniformly across all populations, reflecting the calibrated standing genetic variation prior to the onset of selection by SSWD. Pathogen traits ($T_{\text{vbnc}}$, $v_{\text{local}}$) are likewise initialized at their calibrated 2012 values. No burn-in period is applied; the epizootic itself (2013–2015) serves as the dynamical transition from pre-disease equilibrium to the post-outbreak regime that is the focus of the forecast analysis.

## 4 Results

We present results from the four forecast scenarios described in Section 3. Readers encountering unfamiliar model terminology (e.g., VBNC, force of infection, sweepstakes reproduction) may wish to consult the Model Description (Section 5) for mechanistic details.

Results focus on coast-wide population trajectories (§4.1), regional recovery patterns (§4.2), climate effects (§4.3), evolutionary dynamics (§4.4), and the specific question of how pathogen evolution shapes long-term outcomes (§4.5). All numerical values are three-seed means from the W154 production model.

### 4.1 Population Trajectories

#### 4.1.1 Coast-Wide Dynamics

The coast-wide population trajectory under the baseline scenario (F01) displays three distinct phases (Figure 1). The first phase is the initial catastrophic decline (2013–2015), during which SSWD sweeps through virtually the entire range of *Pycnopodia helianthoides*, reducing coast-wide abundance by more than 90%. This crash is driven by the combination of anomalously warm SSTs during the 2013–2016 northeast Pacific marine heatwave and the high initial susceptibility of host populations ($r_0 = 0.150$).

The second phase (2015–2025) is characterized by intermittent, incomplete recovery attempts. Recruitment pulses—transient surges in juvenile abundance driven by seasonal cooling and residual reproductive output—produce brief periods of apparent population growth, but these gains are systematically erased during subsequent warm seasons when pathogen activity resumes. This "two steps forward, one-and-a-half steps back" dynamic produces a sawtooth population trajectory with a weak positive trend that fails to restore populations to pre-epizootic levels.

The third phase (2025–2050) sees a gradual, secular decline as the cumulative toll of recurring epizootics, progressive warming, and pathogen adaptation erodes the demographic buffer. By 2050, coast-wide abundance under F01 stabilizes at approximately 238,687 individuals, or 5.4% of the pre-disease peak of 4,447,241 (Table 2). The other three scenarios produce strikingly similar

coast-wide outcomes: F02 retains 5.9% (261,632), F03 retains 5.1% (224,638), and F04 retains 5.8% (258,760). The narrow range of these values (5.1–5.9%) belies dramatic regional redistribution, as detailed below.

Table 2: Coast-wide population outcomes at 2050 across the four forecast scenarios. Values are three-seed means.

| Scenario | Survivors | Initial | % Surviving |
|---|---|---|---|
| F01 (SSP2-4.5, full evol.) | 238,687 | 4,447,241 | 5.4 |
| F02 (SSP2-4.5, no evol.) | 261,632 | 4,447,241 | 5.9 |
| F03 (SSP2-4.5, thermal only) | 224,638 | 4,447,241 | 5.1 |
| F04 (SSP5-8.5, full evol.) | 258,760 | 4,447,241 | 5.8 |

#### 4.1.2 Regional Divergence

Beneath the superficially uniform coast-wide totals, regional trajectories diverge markedly (Figure 3). Northern populations (Alaska, British Columbia) display persistent seasonal cycling: abundance builds during winter and spring when SSTs fall below the pathogen's viable-but-non-culturable (VBNC) threshold, then declines during summer and autumn as warming reactivates the pathogen. This oscillatory regime reflects the strong seasonality of high-latitude SSTs and the proximity of ambient temperatures to the pathogen's thermal activation boundary.

In contrast, southern populations (California, Baja California) exhibit monotonic decline with minimal seasonal modulation. Year-round warm SSTs in these regions maintain chronic pathogen activity, leaving populations no seasonal refuge. The result is a relentless erosion of abundance that, in the most southerly regions, proceeds to functional or absolute extinction (Section 4.2).

#### 4.1.3 The Recruitment Pulse Phenomenon

A conspicuous feature of the simulated trajectories is the recurrence of recruitment pulses—brief (2–6 month) surges in abundance that interrupt the long-term decline. These pulses arise when a sequence of cooler-than-average months suppresses pathogen activity long enough for surviving adults to reproduce and juveniles to recruit into the population. However, the pulses are inherently self-limiting: the temporary increase in host density raises the effective contact rate, and the subsequent return to warmer conditions triggers renewed epizootic mortality that rapidly erases the demographic gain.

This dynamic is most pronounced in the mid-latitude transitional zone (Washington, Oregon), where interannual SST variability is high and the population teeters near the pathogen's thermal activation boundary. In Alaska, pulses are smaller in amplitude but more regularly spaced, tracking the predictable annual thermal cycle. In southern California and Baja, recruitment pulses are largely absent by the 2030s, as chronic pathogen pressure suppresses reproduction below replacement (Figure 10).

### 4.2 Regional Recovery Patterns

Table 3 presents the full matrix of regional recovery percentages (surviving population as a fraction of initial abundance) across all 18 model regions and four scenarios.

Table 3: Regional recovery at 2050 (% of initial population surviving), grouped by geographic cluster. Values are three-seed means. Bold values indicate the highest recovery within each region across scenarios.

| Cluster | Region | F01 | F02 | F03 | F04 |
|---|---|---|---|---|---|
| Alaska | AK-AL | 0.4 | **4.1** | 0.7 | 0.4 |
| | AK-WG | 2.5 | **4.5** | 3.0 | 2.4 |
| | AK-OC | 1.8 | 3.1 | 1.5 | **6.0** |
| | AK-EG | 1.7 | **2.4** | 2.0 | 1.6 |
| | AK-PWS | 2.4 | **2.6** | 2.5 | 2.2 |
| | AK-FN | 6.8 | 6.4 | 5.7 | **9.9** |
| | AK-FS | 2.7 | 3.6 | 2.9 | **10.3** |
| BC/SS | BC-N | 3.6 | 4.9 | 3.7 | **11.3** |
| | BC-C | 3.3 | **7.5** | 4.1 | 6.4 |
| | SS-N | 2.4 | **3.6** | 2.6 | 3.1 |
| | SS-S | 6.5 | **8.8** | 7.7 | 7.6 |
| PNW | JDF | 5.4 | **6.5** | 5.9 | 5.3 |
| | WA-O | **8.8** | 5.1 | 4.2 | 6.0 |
| | OR | **17.6** | 12.8 | 15.3 | 10.4 |
| CA/BJ | CA-N | 5.0 | **5.9** | 2.9 | 4.3 |
| | CA-C | **11.9** | 11.9 | 11.5 | 7.6 |
| | CA-S | 0.0 | 0.1 | **0.4** | 0.0 |
| | BJ | 0.0 | 0.0 | 0.0 | 0.0 |

#### 4.2.1 Alaska

Alaskan populations uniformly show poor recovery across all scenarios, with most regions retaining fewer than 3% of initial abundance under the baseline (F01). The exception is AK-FN (Funter Bay–northern Southeast Alaska), which achieves 6.8% under F01 and reaches 9.9% under the high-warming scenario F04. The consistently low values throughout western and south-central Alaska reflect the combination of cold winters that limit the reproductive season and summers that, despite being cool by lower-latitude standards, are now warm enough to sustain episodic pathogen outbreaks.

Notably, several Alaskan regions show their strongest recovery under F04 (SSP5-8.5), particularly AK-FS (+7.6 percentage points relative to F01) and AK-OC (+4.2 points). This counterintuitive result—warming *helping* northern populations—is discussed in Section 4.3.

#### 4.2.2 British Columbia and the Salish Sea

British Columbia and the Salish Sea region display moderate recovery potential (3–9%) with substantial sensitivity to both climate and evolutionary scenarios. BC-N (northern British Columbia) is among the most climate-responsive regions in the entire domain, increasing from 3.6% under F01 to 11.3% under F04—a gain of 7.7 percentage points that makes it the single largest beneficiary of high-end warming (Figure 2). BC-C (central British Columbia) is instead most sensitive to pathogen evolution, jumping from 3.3% (F01) to 7.5% (F02, evolution off), a gain of 4.2 points.

The Salish Sea subregions (SS-N, SS-S) exhibit intermediate recovery and moderate scenario sensitivity. SS-S consistently outperforms SS-N, likely reflecting the greater habitat complexity and

cooler deep-water refugia of the southern basins.

#### 4.2.3 Pacific Northwest: Oregon's Dominance

The Pacific Northwest cluster reveals the most striking regional pattern in the entire forecast ensemble. Oregon (OR) is the unambiguous recovery leader under the baseline scenario, retaining 17.6% of its initial population—more than three times the coast-wide average and nearly double the next-best region (CA-C at 11.9%). Oregon's advantage derives from a combination of factors: strong upwelling that maintains relatively cool summer SSTs, high initial population density that buffers against stochastic extinction, and sufficient thermal variability to provide seasonal windows of low pathogen activity.

However, Oregon is also the region most vulnerable to both warming and evolutionary change. Under F04 (high warming), Oregon's recovery drops to 10.4%—a loss of 7.2 percentage points, the largest negative climate effect of any region. Under F02 (no evolution), Oregon declines to 12.8%, losing 4.8 points despite the coast-wide benefit of freezing pathogen evolution. These paradoxical vulnerabilities are analyzed in Sections 4.3 and 4.5.

Outer Washington (WA-O) follows a distinctive pattern, achieving its highest recovery (8.8%) under F01 but declining under all alternative scenarios—including F02, where disabling pathogen evolution actually reduces its recovery by 3.7 percentage points.

#### 4.2.4 California and Baja California: Southern Collapse

Southern California (CA-S) and Baja California (BJ) are functionally extinct across all scenarios. CA-S retains at most 0.4% of its initial population (under F03), while BJ achieves 0.0% in every scenario. These results indicate that even under moderate warming with favorable evolutionary assumptions, the southern range limit of *P. helianthoides* has contracted northward past approximately 33°N and is unlikely to recover within the forecast horizon.

Central California (CA-C) presents a more nuanced picture: it maintains robust recovery (11.9%) under both F01 and F02 but declines substantially (to 7.6%) under high warming (F04). Northern California (CA-N) is intermediate, with moderate recovery (5.0–5.9%) under SSP2-4.5 scenarios but reduced persistence (4.3%) under SSP5-8.5.

The extinction of southern populations represents the loss of a significant fraction of the species' historical genetic diversity and raises conservation concerns that extend beyond simple abundance metrics (Section 4.4).

### 4.3 Climate Effects

The contrast between F01 (SSP2-4.5) and F04 (SSP5-8.5), both with full evolutionary dynamics, isolates the effect of enhanced ocean warming on population outcomes. Coast-wide, the net effect is remarkably small: F04 retains 5.8% of the initial population versus 5.4% under F01, a difference of only +0.4 percentage points. This near-zero aggregate effect, however, masks a dramatic latitudinal redistribution of surviving populations (Figure 4).

#### 4.3.1 Northern Beneficiaries

The largest positive climate effects are concentrated in a band from southern Alaska through northern British Columbia (Figure 9):

- BC-N: +7.7 percentage points (3.6% → 11.3%)
- AK-FS: +7.6 points (2.7% → 10.3%)
- AK-OC: +4.2 points (1.8% → 6.0%)
- AK-FN: +3.1 points (6.8% → 9.9%)
- BC-C: +3.1 points (3.3% → 6.4%)

The mechanism underlying this northern benefit operates through two complementary pathways. First, warming extends the effective growing season for *P. helianthoides* at high latitudes, where cold-limited reproduction constrains population growth rates more severely than disease-induced mortality during most of the year. Second, although warming also extends the pathogen's active season, the absolute SSTs in these regions remain near the lower boundary of the pathogen's thermal niche even under SSP5-8.5, limiting the intensity and duration of summer epizootics. The net balance thus favors the host: more reproduction in spring and autumn, with only modestly increased disease pressure in summer.

#### 4.3.2 Southern Losers

South of approximately 48°N, warming uniformly depresses recovery:

- OR: −7.2 percentage points (17.6% → 10.4%)
- CA-C: −4.3 points (11.9% → 7.6%)
- WA-O: −2.8 points (8.8% → 6.0%)
- CA-N: −0.7 points (5.0% → 4.3%)

In these regions, baseline SSTs are already warm enough to sustain prolonged pathogen activity. Additional warming expands the window of active transmission into what were previously marginal shoulder seasons, increasing the cumulative annual disease burden. Simultaneously, these populations are already near the upper thermal tolerance for reproduction, so warming provides little compensatory demographic benefit. Oregon suffers the steepest absolute decline because it has the most to lose: its high baseline recovery rate amplifies the proportional impact of extended pathogen activity seasons.

#### 4.3.3 The Breakeven Latitude

The transition from net climate benefit (north) to net climate cost (south) occurs at approximately 48°N—roughly the latitude of the Juan de Fuca Strait and the boundary between the upwelling-dominated California Current system and the downwelling regime of the northern shelf. This breakeven latitude is not merely a statistical artifact but reflects a genuine biophysical boundary: north of 48°N, winter SSTs are sufficiently cold that the pathogen enters prolonged VBNC dormancy, and warming's primary effect is to accelerate host population growth during the expanded ice-free season. South of this boundary, the pathogen's dormancy period is already short or absent, and warming's primary effect is to intensify chronic disease pressure.

The existence of this breakeven latitude has important conservation implications. Under high-end warming, the center of abundance for *P. helianthoides* shifts northward by several degrees of latitude, with northern British Columbia and southeastern Alaska emerging as the primary refugia. This geographic shift could alter the species' ecological role along the coast, as regions that currently

benefit from sunflower star predation on sea urchins (notably the California kelp forests) would lose this keystone function while regions with less acute urchin overgrazing pressure gain it.

## 4.4 Evolutionary Dynamics

The coupled eco-evolutionary framework tracks the temporal evolution of both host resistance and pathogen traits at each of the 896 model sites. Under the baseline scenario (F01), these traits exhibit striking spatial gradients and temporal dynamics that illuminate the interplay between selection, drift, and demographic constraint.

### 4.4.1 Host Resistance

Mean host resistance ($\bar{r}$) increases from the uniform initial value of 0.150 toward higher values in all surviving populations, but the rate and ultimate level of resistance evolution vary enormously across the latitudinal range (Figure 5):

- **Alaska (moderate selection, slow response):** AK-PWS: 0.150 → 0.253; AK-FN: 0.150 → 0.230. Disease pressure is seasonal and moderate, producing consistent but gentle directional selection.
- **British Columbia (intermediate):** BC-N: 0.150 → 0.240. Stronger summer epizootics drive faster allele frequency change than in Alaska, but demographic stability maintains sufficient population sizes for effective selection.
- **Oregon and California (intense selection, rapid response):** OR: 0.150 → 0.378; CA-C: 0.150 → 0.425. Year-round pathogen activity imposes sustained, intense selection for resistance. CA-C achieves the highest resistance of any surviving population, with mean $\bar{r}$ reaching 0.43 by 2050.
- **Southern California (extinction):** CA-S: 0.150 → 0.115 (approximately 52 survivors). Mean resistance actually *declines* below the starting value, because the population is so small (∼50 individuals) that genetic drift overwhelms directional selection.

A critical observation is that even the highest evolved resistance level ($\bar{r} = 0.425$ in CA-C) provides only partial protection: at this resistance level, 58% of the force of infection still penetrates host defenses. In the context of the model's multiplicative disease framework, this means that CA-C populations in 2050 still experience more than half the per-capita disease mortality of a completely naïve population. Resistance evolution slows the decline but does not arrest it.

### 4.4.2 The Cruel Paradox of Evolutionary Capacity

The populations under the most intense selection for resistance are precisely those with the least capacity to respond. In southern populations, chronic pathogen pressure drives rapid allele frequency change but simultaneously erodes population size, reducing the effective population size ($N_e$) and hence the efficiency of selection. As $N_e$ declines, genetic drift increasingly overwhelms directional selection, and the additive genetic variance available for further adaptation collapses (Figure 6).

This evolutionary trap is starkest in CA-S, where the population declines to approximately 52 individuals by 2050 despite experiencing intense selection pressure. Remarkably, the mean resistance of these survivors ($\bar{r} = 0.115$) is actually *lower* than the starting value (0.150)—evidence that in such a tiny population, genetic drift has overwhelmed directional selection. The few surviving individuals are not the "most resistant"; they are simply the ones that happened to survive through

stochastic chance. Allee effects and demographic stochasticity now dominate their fate far more than any genetic advantage.

The paradox generalizes across the southern range: the populations that "need" evolutionary rescue the most are those least likely to achieve it, because the disease that creates the selection pressure also destroys the demographic base required for effective evolutionary response. This result echoes theoretical predictions from evolutionary rescue theory [Gomulkiewicz and Holt, 1995, Bell, 2008] and provides a concrete, spatially explicit demonstration of its consequences for a real species undergoing an ongoing epizootic.

#### 4.4.3 Pathogen Thermal Adaptation

The pathogen's VBNC threshold temperature ($T_{\text{vbnc}}$)—the SST below which the pathogen enters dormancy—evolves toward lower values in cold-water regions as strains with broader thermal niches gain a fitness advantage through extended transmission seasons. By 2050 under F01, $T_{\text{vbnc}}$ has declined to its implemented lower bound of 9.00°C throughout Alaska (AK-PWS: 9.00°C; AK-FN: 9.00°C), indicating that selection has driven the trait to the edge of the modeled parameter space within 38 years (Figure 7).

At lower latitudes, $T_{\text{vbnc}}$ settles at intermediate values that reflect a balance between selection for broader thermal niches and the diminishing marginal fitness return of further thermal adaptation in already-warm environments:

- BC-N: $T_{\text{vbnc}} = 9.39$°C
- OR: $T_{\text{vbnc}} = 10.70$°C
- CA-C: $T_{\text{vbnc}} = 11.81$°C
- BJ: $T_{\text{vbnc}} = 12.00$°C (upper bound, unchanged from initialization)

The latitudinal gradient in $T_{\text{vbnc}}$ is opposite to what might naïvely be expected. In warm regions, the pathogen has no "need" to adapt its thermal niche because ambient temperatures already exceed activation thresholds year-round; consequently, $T_{\text{vbnc}}$ remains near its initial value. In cold regions, even small reductions in $T_{\text{vbnc}}$ substantially extend the annual transmission window, creating strong selection for thermal generalism.

The saturation of $T_{\text{vbnc}}$ at 9.00°C throughout Alaska by 2050 implies that pathogen thermal adaptation has effectively reached its maximum extent in the north within the modeled range. Further increases in pathogen pressure at high latitudes would require either warming of SSTs above the adapted threshold for longer periods or evolution of $T_{\text{vbnc}}$ below the 9°C floor—a possibility our model does not permit but that merits investigation in future work.

#### 4.4.4 Virulence Gradient

Local virulence ($v_{\text{local}}$) evolves independently at each site, driven by the interplay between within-host selection for increased pathogen replication and the epidemiological cost of host mortality. By 2050, a pronounced latitudinal virulence gradient has developed (Figure 7):

| **Region** | $v_{\text{local}}$ (2050) |
|---|---|
| AK-PWS | 0.081 |
| AK-FN | 0.091 |
| BC-N | 0.144 |
| OR | 0.138 |
| CA-C | 0.235 |
| BJ | 0.335 |

Virulence increases fourfold from Alaska (0.086) to Baja California (0.343). This gradient arises because southern populations sustain chronic, dense pathogen populations year-round, favoring aggressive strains that maximize short-term transmission. In the north, the annual interruption of transmission by winter cooling selects against hyper-virulent strains that kill their hosts before the next transmission season—a classic prediction of the transmission–virulence trade-off hypothesis applied to a seasonally structured environment [Alizon et al., 2009].

The virulence gradient reinforces the north–south asymmetry already created by climate: southern populations face not only longer pathogen activity seasons but also intrinsically more virulent pathogen populations, compounding the demographic disadvantage.

## 4.5 The Pathogen Evolution Question

A central motivation of this study is to determine whether pathogen evolution materially alters the long-term forecast for *P. helianthoides*. The factorial design permits direct quantification of this effect through comparison of scenarios with (F01) and without (F02) pathogen evolutionary capacity, and through the intermediate case where only virulence evolution is disabled (F03).

### 4.5.1 Aggregate Effect: Modest and Counterintuitive

Coast-wide, disabling all pathogen evolution (F02) yields a modest improvement: 5.9% survival versus 5.4% under F01, a difference of +0.5 percentage points (Figure 8). This result confirms the intuition that pathogen adaptation modestly worsens host population outcomes, but the small magnitude is surprising given the substantial trait changes documented in Section 4.4. Two features of the regional breakdown are particularly unexpected.

### 4.5.2 Where Disabling Evolution Hurts the Host

Counterintuitively, removing pathogen evolution *reduces* recovery in Oregon (−4.8 percentage points: 17.6% → 12.8%) and outer Washington (−3.7 points: 8.8% → 5.1%). These are among the most important recovery regions under the baseline, and the loss is substantial.

The mechanism driving this paradox involves the interaction between thermal adaptation and virulence evolution. When pathogen evolution is enabled (F01), the pathogen's VBNC threshold in the Pacific Northwest declines from its initial value of 12.0°C toward ∼10.6°C (Oregon), extending the pathogen's active season by several weeks. However, this same evolutionary process simultaneously allows virulence to drift *downward* from its initial value toward a local optimum dictated by the transmission–virulence trade-off. In Oregon, evolved virulence (0.136) is substantially below the pathogen's initial virulence parameterization, meaning that the pathogen becomes less lethal per transmission event even as it becomes more broadly active across seasons.

When evolution is disabled (F02), the pathogen retains its initial $T_{\text{vbnc}}$ of 12.0°C, which in Oregon's thermal regime restricts activity to a shorter summer window. Crucially, however, the pathogen also retains its initial virulence, which exceeds the locally optimal value. In the Pacific Northwest's thermal regime, where winter cooling provides seasonal population recovery, the combination of a shorter but more virulent active season (F02) proves *more damaging* than a longer but less virulent season (F01). The elevated per-encounter lethality during the concentrated summer peak overwhelms the benefit of a shorter overall transmission window.

This result has important biological implications: the net effect of pathogen evolution on host populations is not uniformly negative. In thermally seasonal environments, the co-evolution of thermal niche and virulence can produce a regime in which the pathogen becomes "gentler but more persistent"—a combination that, paradoxically, allows greater host population recovery than the alternative of "fierce but brief" epizootics.

#### 4.5.3 Where Disabling Evolution Helps the Host

In most other regions, particularly at high latitudes, removing pathogen evolution benefits the host. The largest gains under F02 occur in:

- BC-C: +4.2 points (3.3% → 7.5%)
- AK-AL: +3.7 points (0.4% → 4.1%)
- AK-WG: +2.0 points (2.5% → 4.5%)
- SS-S: +2.3 points (6.5% → 8.8%)

In these cold-water regions, the dominant evolutionary axis is pathogen thermal adaptation (downward movement of $T_{\text{vbnc}}$), which extends the pathogen's active season into previously safe shoulder months. Unlike in the Pacific Northwest, the virulence trade-off provides little compensatory benefit here because the short active season limits the number of transmission events over which reduced virulence can accumulate a demographic advantage. Consequently, disabling evolution simply preserves the shorter pre-adaptation activity window without offsetting virulence costs.

#### 4.5.4 Virulence Evolution Alone: A Minor Player

Comparison of F01 (full evolution) with F03 (thermal adaptation on, virulence off) isolates the specific contribution of virulence evolution. The effect is negligible: coast-wide survival changes from 5.4% (F01) to 5.1% (F03), a difference of only −0.3 percentage points. Regionally, the largest effects of disabling virulence evolution alone are:

- CA-N: −2.1 points (5.0% → 2.9%)
- OR: −2.3 points (17.6% → 15.3%)
- WA-O: −4.6 points (8.8% → 4.2%)
- AK-FN: −1.1 points (6.8% → 5.7%)

These modest, mostly negative effects (i.e., disabling virulence evolution slightly *worsens* outcomes) are consistent with the mechanism described above: virulence evolution tends to reduce per-encounter lethality below initial values, providing a small net benefit to host populations. When this beneficial drift is prevented (F03), outcomes are slightly worse.

Importantly, the contrast between the large effect of disabling *all* evolution (F02 vs. F01) and the small effect of disabling virulence evolution alone (F03 vs. F01) implies that **thermal adaptation is the dominant evolutionary axis** driving the pathogen's impact on host populations. Virulence evolution, while biologically real and spatially structured, is a secondary modulator whose effects are largely masked by the much larger influence of the pathogen's expanding thermal niche.

#### 4.5.5 Synthesis: The Evolutionary Forecast

The evolutionary analysis yields three key findings:

1. **Pathogen evolution has a modest net negative effect on host populations** ($-0.5\%$ coast-wide), but this aggregate conceals regionally divergent impacts that range from $+4.8\%$ (Oregon benefits from evolution) to $-4.2\%$ (BC-C harmed by evolution).
2. **Thermal adaptation, not virulence evolution, is the primary driver.** The pathogen's ability to expand its thermal niche and extend its active season is far more consequential than changes in per-encounter virulence.
3. **The host's evolutionary response (increasing resistance) is substantial but insufficient.** Even the most resistant surviving population (CA-C, $\bar{r} = 0.425$) still experiences more than half the disease mortality of naïve animals, and the populations most in need of evolutionary rescue are those least able to achieve it due to demographic collapse.

These results argue against a simple narrative of pathogen evolution as uniformly detrimental and instead point to a complex, spatially structured eco-evolutionary dynamic in which the net effect of evolution depends critically on the local thermal regime, the seasonal structure of transmission, and the interaction between thermal niche expansion and virulence optimization.

# 5 Model Description

This section describes the structure and mechanics of the SSWD-EvoEpi agent-based model. Readers interested primarily in conservation implications may proceed to the Discussion (Section 6); what follows is intended for those who wish to understand how the forecasts were generated.

## 5.1 Agent-Based Framework

The model simulates individual *Pycnopodia helianthoides* across **896 coastal sites** spanning the species' entire historical range, from the western Aleutian Islands (52.5°N) to central Baja California (28.5°N). Sites are grouped into 18 geographic regions corresponding to major oceanographic and biogeographic boundaries (e.g., Prince William Sound, Salish Sea, Oregon coast, Channel Islands). Each site supports up to $K = 5{,}000$ individuals, yielding a theoretical coast-wide carrying capacity of approximately 4.5 million sea stars.

The simulation operates in **daily timesteps** from 2012 through 2050. Each simulated day steps through disease transmission, individual survival, growth, and—during the spawning season—reproduction and larval dispersal.

Each individual sea star is represented as a distinct agent carrying the following attributes:

- **Position** ($x, y$ in meters) and **heading** (direction of travel) within its home site;

- **Body size** (arm-tip diameter in mm, growing via von Bertalanffy dynamics toward $L_\infty = 1{,}000\,\text{mm}$);
- **Age** and **sex**;
- **Disease state** (susceptible, exposed, pre-symptomatic, symptomatic, or dead);
- **51 genetic loci** governing three heritable defense traits (Section 5.3).

Individuals move within sites via a **correlated random walk** at a base speed of 0.5 m/min [Hodin et al., 2021], resolved in 24 hourly substeps per day. Symptomatic individuals slow to 10% of normal speed; dead individuals do not move. Movement matters because disease transmission is **spatially explicit**: a 20 m grid within each site tracks local pathogen concentrations, so stars near clusters of infected individuals experience higher exposure than those farther away.

**Temperature forcing.** All scenarios are driven by **observed sea surface temperature (SST)** from the NOAA Optimum Interpolation SST v2.1 dataset for 2012–2025, bilinearly interpolated to each of the 896 model sites. For the projection window (2026–2050), SSTs are generated via a delta method applied to CMIP6 general circulation models (Section 5.7). Real climate events—including the 2013–2016 marine heatwave, El Niño, and La Niña cycles—are thus automatically incorporated during the historical period.

## 5.2 Disease Mechanics

The disease module simulates the dynamics of *Vibrio pectenicida*, recently confirmed as the etiological agent of SSWD through fulfillment of Koch's postulates [Prentice et al., 2025]. Its structure rests on two biological pillars: a modified SEIR compartmental model within individual hosts, and the VBNC (viable but non-culturable) dormancy state that couples pathogen activity to water temperature.

**Within-host progression.** Each individual follows a five-stage disease progression:

$$\text{S} \rightarrow \text{E} \rightarrow \text{I}_1 \text{ (pre-symptomatic)} \rightarrow \text{I}_2 \text{ (symptomatic)} \rightarrow \text{Dead}$$

Transition rates between compartments are Arrhenius-scaled (i.e., exponentially increasing with temperature) to local SST, meaning disease progresses faster in warmer water. At a reference temperature of 20°C, the mean time from exposure to death is approximately 6 days ($\mu_{EI_1} = 0.233/\text{d}$, $\mu_{I_1 I_2} = 0.434/\text{d}$, $\mu_{I_2 D} = 0.563/\text{d}$). Pre-symptomatic individuals shed pathogen at a rate of $\sigma_1 = 5.0$ units/day; symptomatic individuals shed ten-fold more ($\sigma_2 = 50.0$). Dying stars release a burst of $\sigma_D = 15.0$ units. Recovered individuals return to susceptible status—echinoderms lack adaptive immunity, so reinfection is possible [Prentice et al., 2025].

**The VBNC state.** *Vibrio* species enter a dormant VBNC state in cold water [Liu et al., 2019]. We model this as a sigmoid transition centered at $T_{\text{vbnc}} = 12°\text{C}$ (steepness $k = 2.0$): at 14°C the pathogen is ∼98% active, at 12°C it is 50% active, and at 10°C only ∼2% remains active. A biophysical floor of $T_{\text{vbnc,min}} = 9°\text{C}$ prevents any evolutionary adaptation below this temperature (Section 4.5).

This mechanism has profound geographic consequences. In southern California, where SST routinely exceeds 12°C, the pathogen is active year-round and disease pressure is relentless. In Alaska,

where SST only briefly exceeds the threshold in summer, stars receive a seasonal reprieve during the cold months—but the pathogen persists in dormancy, ready to reactivate each warm season.

**Environmental pathogen pool.** Each site maintains a dynamic environmental reservoir of *Vibrio*. The pool grows as infected individuals shed pathogen (fraction $\alpha_{\text{env}} = 0.18$ enters the persistent pool) and decays naturally ($\delta_{\text{env}} = 0.02$/day, half-life ≈35 days). A susceptible star's daily infection probability follows Michaelis–Menten kinetics with a half-saturation constant $K_{\text{half}} =$ 800,000, modified by the individual's resistance trait, body size, and local VBNC activation. This density-dependent mechanism creates a positive feedback loop: more infected individuals shed more pathogen, raising the environmental pool, which drives further infections. At high host density, this feedback can collapse a site within weeks; at low density, the same parameters produce modest mortality because the pool never builds to saturating levels.

**Wavefront propagation.** Consistent with observational records, disease originates from five sites in the Channel Islands (southern California) in mid-2013 and spreads as a northward wave. Three mechanisms interact to produce the wavefront: (1) thermal activation requires SST to exceed $T_{\text{vbnc}}$ before the pathogen can establish locally, so southern sites ignite first; (2) a cumulative dose threshold of 1,000 units must be reached before local outbreak onset, creating a realistic lag between first exposure and epizootic; and (3) waterborne pathogen disperses between connected sites via an exponential decay kernel ($D_P = 300\,\text{km}$ characteristic distance), allowing the disease front to propagate along the coast.

### 5.3 Host Genetics and Evolution

Each individual carries **51 diploid genetic loci** divided equally among three heritable defense traits, inspired by genome-wide association study (GWAS) results from the congener *Pisaster ochraceus* [Schiebelhut et al., 2018]—the only Pacific asteroid for which disease-associated genomic data currently exist. No equivalent GWAS has yet been conducted for *Pycnopodia*.

**Resistance ($r$, 17 loci):** Scales the individual's infection hazard rate by $(1 - r)$. A star with $r = 0.30$ experiences a 30% reduction in its instantaneous infection risk per unit pathogen exposure. Think of it as an *armor rating*.

**Tolerance ($t$, 17 loci):** Extends survival during symptomatic ($I_2$) infection, reducing the rate of progression to death. A tolerant star lives longer while infected, buying time for possible clearance.

**Recovery ($c$, 17 loci):** Scales the daily probability of clearing the pathogen. Effective daily clearance is $\rho_{\text{rec}} \times c$, where $\rho_{\text{rec}} = 0.05$. Cleared individuals return to susceptible status (R→S; no adaptive immunity in echinoderms).

Locus effect sizes follow an exponential distribution (a few loci of large effect, many of small effect), and traits are computed as additive sums across their respective loci. Initial trait values—$r_0 = 0.15$, $t_0 = 0.10$, $c_0 = 0.02$—reflect a largely naïve population prior to the 2013 outbreak. At $c_0 = 0.02$, the effective daily clearance rate is just 0.1%, meaning recovery from active infection is extremely rare.

**Natural selection.** Selection operates straightforwardly: individuals that survive disease disproportionately carry favorable alleles at defense loci. When survivors reproduce, they pass those alleles to offspring. Over generations, mean defense trait values increase. By 2050, resistance in

hard-hit southern populations rises from 0.15 to approximately 0.43, while less-affected northern populations evolve more slowly ($r \approx 0.20$–$0.25$). However, even $r = 0.45$ still leaves more than half of the pathogen force unblocked—evolution is real but not a rescue.

A critical secondary effect is **genetic diversity loss**. Populations that crash to near-zero lose most of their standing genetic variation. The additive genetic variance ($V_A$) for resistance in southern California—the raw material for future adaptation—collapses from 0.026 to 0.003, an 88% reduction. The populations most in need of continued evolution have the least capacity for it.

### 5.4 Pathogen Community Evolution

Rather than tracking individual bacterial strains, the model treats the pathogen community at each site as the evolutionary unit—an aggregate whose properties shift in response to local selection pressures. Each community has two evolving traits:

**Thermal activation threshold ($T_{\text{vbnc}}$).** The temperature midpoint for the VBNC sigmoid drifts downward at cold-water sites where there is sustained selection for cold-tolerant strains. In Alaska, the community threshold evolves from 12°C to approximately 9.0–9.4°C over the simulation, extending the active disease season by several weeks. At warm southern sites, where the pathogen is already active year-round, the threshold barely changes—there is no selection pressure for further cold adaptation. Evolution of this trait is bounded by a hard biophysical floor at 9°C.

**Local virulence ($v_{\text{local}}$).** Virulence evolves according to the classic virulence–transmission trade-off [Anderson and May, 1982]. A more virulent pathogen kills hosts faster, generating larger pathogen bursts, but also terminates shedding sooner. The optimal strategy depends on local conditions: in warm, high-density environments, high virulence is favored (quick kill, many available hosts); in cold, sparse environments, lower virulence is favored (host kept alive longer, maximizing total transmission during a brief warm window). Each site's community drifts toward a local optimum, producing a latitudinal gradient from $v \approx 0.08$ in Alaska to $v \approx 0.35$ in Baja California.

The combination of thermal adaptation and virulence evolution creates distinct pathogen strategies across the range: northern communities are *more cold-tolerant but less virulent*, while southern communities are *already warm-adapted but highly aggressive*.

### 5.5 Reproduction and Dispersal

**Broadcast spawning.** Sunflower stars reproduce via broadcast spawning, releasing eggs and sperm into the water column. Spawning timing follows a latitude-dependent schedule (peak spawning day $= 50 + 3 \times [\text{latitude} - 48.5°]$), with the season running from November through March [Hodin et al., 2021]. Alaskan populations spawn later (mid-March) than Californian populations (mid-January).

**Sweepstakes reproductive success.** Reproductive output follows the Hedgecock sweepstakes hypothesis [Hedgecock and Pudovkin, 2011]: in any spawning event, a small fraction of adults contribute disproportionately to the next generation. We implement this as a Pareto-weighted random lottery (shape $\alpha = 1.35$), where each reproductive adult receives a random weight from a heavy-tailed distribution. This amplifies genetic drift, reduces effective population size ($N_e$) well below the census count, and means that recovery genetics depend heavily on *which* individuals happen to win the reproductive lottery—not just how many survive.

**Larval dispersal.** Larvae disperse between connected sites via an exponential decay kernel with characteristic distance $D_L = 400\,\text{km}$. Rather than modeling hydrodynamic transport, we use a simplified connectivity scheme weighted by overwater distance between sites and modified by coastal geometry. Settler survival is 0.1%—consistent with the extremely high larval mortality typical of marine invertebrates.

**Allee effects.** The model incorporates a **double Allee effect** at depleted sites. First, fertilization success depends on local adult density via Lundquist–Botsford kinetics ($F(\rho) = 1 - e^{-\gamma\rho_{\text{eff}}}$, $\gamma = 4.5\,\text{m}^2$): at low density, eggs fail to encounter sperm. Second, larval settlement success follows a Michaelis–Menten curve from 20% (substrate cues only) to 100% (full adult chemical cues), so sites with few adults are difficult for larvae to recolonize. Together, these mechanisms create an extinction vortex at very low densities—directly relevant to reintroduction planning, where release cohorts must be large enough to overcome both thresholds.

Pre-spawning aggregation behavior partially mitigates the fertilization Allee effect: for 14 days before and after spawning, individuals' movement is biased toward nearby conspecifics within a 100 m detection range, raising effective local density in clumps.

**The enclosedness metric.** A novel feature of the model is an algorithmic measure of each site's geographic enclosure based on surrounding coastline geometry. Sites deep within fjords or inland seas score high; exposed outer-coast headlands score low. Enclosedness controls an inherent trade-off:

- **Larval retention** increases at enclosed sites (good for recovery—local reproduction contributes more to local recruitment).
- **Pathogen retention** also increases (bad for disease escape—the same physical barriers that trap larvae trap waterborne *Vibrio*).

Per-site flushing rates are computed as $\varphi = f(\text{enclosedness},\ n_{\text{conn}} = 0.3)$, with the nonlinear exponent $n_{\text{conn}}$ calibrated to balance these opposing effects. Self-recruitment fractions interpolate continuously from 2% (open coast) to 70% (deep fjord).

## 5.6 Calibration Summary

The production model (designated **W154**) was selected after 154 iterative calibration rounds, each testing different parameter combinations against empirical data. Two classes of targets were used:

**Disease arrival timing:** 17 regional targets for when SSWD was first documented, from the Channel Islands (mid-2013) to Alaska (2015–2016).

**Recovery fractions:** Fractional population recovery at each region by 2024, based on the best available survey data [Gravem et al., 2021, Gehman et al., 2025]. Targets span four orders of magnitude: 50% for Prince William Sound (substantial cold-water recovery) down to 0.1% for northern California (near-extirpation in warm water).

Overall fit is quantified by the **log-RMSE** (root-mean-square error of log-transformed recovery fractions), which weights relative errors equally across the four-order-of-magnitude range of targets. W154 achieves a three-seed mean RMSE of **0.504** (best single seed: 0.479).

Key calibrated parameters:

| Parameter | Value | Interpretation |
|---|---|---|
| $K$ | 5,000 | Per-site carrying capacity |
| $K_{\text{half}}$ | 800,000 | Environmental pathogen for 50% infection rate |
| $\alpha_{\text{env}}$ | 0.18 | Fraction of host shedding entering persistent pool |
| $\delta_{\text{env}}$ | 0.02/d | Environmental pathogen decay rate |
| $n_{\text{conn}}$ | 0.3 | Enclosedness-to-flushing nonlinearity |
| $D_P$ | 300 km | Waterborne pathogen dispersal distance |
| $D_L$ | 400 km | Larval dispersal distance |

**Known limitation.** The model's main calibration challenge is Alaska. W154 produces 2–7% recovery in Prince William Sound and Fairweather against 50% survey-based targets. This tension reflects a fundamental model constraint: strengthening the cold-water reprieve enough to allow 50% Alaskan recovery tends to also permit unrealistic recovery at mid-latitude sites. We note this transparently as the primary axis of model uncertainty and discuss its implications in Section 6.

### 5.7 SST Projections

Future temperatures (2026–2050) are generated via a **delta method** applied to CMIP6 ensemble projections:

$$T_{\text{site}}(y, m) \;=\; T_{\text{site,clim}}(m) \;+\; \left[\, T_{\text{ref,proj}}(y, m) - T_{\text{ref,baseline}}(m) \,\right] \tag{1}$$

where $T_{\text{site,clim}}(m)$ is the site's observed monthly climatology (2012–2025 mean), $T_{\text{ref,proj}}(y, m)$ is the CMIP6 projection at the nearest reference site, and $T_{\text{ref,baseline}}(m)$ is the corresponding CMIP6 historical baseline. Eleven CMIP6 reference sites distributed at regular latitudinal intervals map onto the 896 model sites via nearest-neighbor assignment. A stochastic daily perturbation term calibrated to observed interannual SST variance is superimposed to maintain realistic thermal variability.

Two climate pathways are used:

**SSP2-4.5 (moderate):** Coast-wide mean warming of approximately +0.8°C by 2050 relative to the 2012–2020 baseline, with stronger warming in the Gulf of Alaska and weaker warming along the upwelling-dominated California Current.

**SSP5-8.5 (high):** Approximately +1.5°C coast-wide, with pronounced high-latitude amplification.

This approach preserves each site's local temperature character (seasonal cycle, interannual variability) while applying the secular warming trend from the CMIP6 ensemble. Its main limitation is spatial coarseness: 11 reference points cannot capture fine-scale features such as local upwelling intensification or freshwater-driven thermal anomalies in fjords.

## 6 Discussion

### 6.1 Recovery without intervention

Across all four forecast scenarios, the model predicts that *Pycnopodia helianthoides* populations will stabilize at approximately 94–95% below pre-disease levels through mid-century (F01: 94.6% decline; F04: 94.2% decline). No combination of climate trajectory and evolutionary capacity

produces meaningful coast-wide recovery to pre-disease abundance. The absolute population floor shifts by tens of thousands of individuals between scenarios, but the qualitative conclusion is unchanged: the species' new normal is a small fraction of its former abundance.

The mechanism underlying this result is what we term the **seasonal disease ratchet**. During winter, falling water temperatures push *V. pectenicida* into VBNC dormancy, disease pressure abates, and surviving adults reproduce. Larval recruitment partially replenishes depleted sites. But each summer, rising temperatures reactivate the pathogen, the environmental pool rebuilds through the shedding–infection positive feedback, and newly recruited cohorts are culled alongside the survivors. The net effect is a population that oscillates around a low equilibrium, gaining recruits in winter and losing them in summer, unable to escape the ratchet. This pattern is especially pronounced from Washington southward, where summer SST consistently exceeds the VBNC activation threshold and disease activity spans 6–10 months of the year.

### 6.2 The Oregon opportunity

Oregon emerges as the most promising region for natural recovery in the model, retaining 17.6% of pre-disease carrying capacity under the baseline scenario (F01)—the highest of any region coast-wide. Three features converge to favor Oregon:

1. **Temperature regime.** Oregon's SST occupies a Goldilocks zone—warm enough that the growing and reproductive season is adequate for population maintenance, but cool enough that the pathogen enters dormancy for several months each winter. The final pathogen thermal threshold at Oregon sites ($T_{\text{vbnc}} \approx 10.7°\text{C}$) reflects moderate cold-adaptation, but winter SST still drops well below this value, providing genuine disease respite.

2. **Open-coast flushing.** Oregon's exposed outer coast scores low on the enclosedness metric, yielding high flushing rates that continuously dilute the environmental pathogen pool. Unlike the semi-enclosed waters of Puget Sound or British Columbia's fjords, where pathogen concentrations build in poorly flushed basins, Oregon's sites experience continuous hydrodynamic renewal.

3. **Moderate virulence.** The local pathogen community has evolved to intermediate virulence ($v_{\text{local}} \approx 0.14$), low enough that hosts survive long enough to reproduce between outbreaks, but high enough that the disease exerts meaningful selection for resistance. Oregon populations have evolved resistance to $r \approx 0.38$ by 2050—high enough to meaningfully slow infection acquisition.

Under high-end warming (SSP5-8.5, F04), Oregon's advantage diminishes (recovery drops to 10.4%) because warmer summers extend the pathogen activity window. This underscores that Oregon's favorable position is thermally mediated and climate-contingent. If conservation managers are seeking sites where natural recovery is most advanced and monitoring is most likely to detect positive trends, Oregon merits particular attention.

### 6.3 Alaska's conditional optimism

Alaska presents a more complicated picture. The model predicts only 2–7% recovery for Alaskan regions under both climate scenarios, but we know this is an underestimate: field surveys document substantially higher recovery in Prince William Sound (estimated ∼50%; [Gehman et al., 2025]). The model's Alaskan underprediction is its single largest calibration shortcoming (Section 5.6) and likely reflects processes not fully captured in the current framework—including freshwater surface

lenses that suppress *Vibrio* in fjord environments, localized upwelling, or higher background cold-tolerance in northern host genotypes.

Setting aside the absolute numbers, the model's *directional* prediction for Alaska is noteworthy. Under SSP5-8.5, Fairweather North recovery increases from 6.8% (F01) to 9.9% (F04)—a 45% relative improvement. Warming extends the growing season and reproductive window in cold-water Alaskan habitats without proportionally extending the disease season, because SST in these regions remains near or below the VBNC floor for much of the year even under high emissions. If real Alaskan populations are indeed doing substantially better than the model shows, then climate change—paradoxically—may create further opportunity for northern populations by expanding favorable thermal habitat.

This has a practical implication: Alaskan populations may be the species' best long-term reservoir, and monitoring investments in Prince William Sound and the Fairweather coast are well justified.

### 6.4 Southern California as a conservation priority

At the opposite end of the range, the model delivers an unambiguous verdict. Southern California (CA-S) and Baja California retain effectively zero population across all scenarios (CA-S: 0.01% recovery, ∼52 individuals coast-wide; Baja: functionally extinct with zero individuals). Central California (CA-C) retains ∼12% under moderate warming but declines to 7.6% under SSP5-8.5, and this trend is downward through 2050 with no indication of stabilization.

Year-round pathogen activity, high evolved virulence ($v_{\text{local}} \approx 0.30$), and the absence of any seasonal disease respite make natural recovery effectively impossible in these regions. The environmental pathogen pool never decays to levels low enough for naïve recruits to survive their first summer. Even with strong selection (resistance evolves to $r \approx 0.43$–$0.45$ in CA-C by 2050), the host cannot evolve fast enough to escape the pathogen's demographic trap.

These findings motivate the upcoming F05–F08 reintroduction scenarios, which will test whether captive-bred juveniles with elevated resistance can establish self-sustaining populations in southern waters. Our results suggest that any reintroduction effort south of Point Conception will require *sustained* releases over multiple years, because single-cohort releases will likely be overwhelmed by the ambient disease pressure before the population can reach densities sufficient to buffer annual losses.

### 6.5 The evolutionary dead end

One of the most striking—and sobering—results is that natural selection, while clearly operating, is insufficient to rescue *Pycnopodia* from SSWD. Host resistance evolves substantially in hard-hit populations: central California's mean resistance increases from $r = 0.15$ to $r = 0.43$ over 38 simulated years, nearly tripling the individual-level hazard reduction. Yet $r = 0.43$ still means that 58% of the pathogen force of infection penetrates the host's defenses. In a high-pathogen environment, this translates to effective daily infection probabilities that remain well above 20%. Resistance buys time, not survival.

The fundamental asymmetry is temporal. *V. pectenicida* can adapt its thermal tolerance and virulence on daily-to-seasonal timescales through community-level selection among trillions of bacteria. The host evolves its defense traits once per generation, with a minimum generation time of several years and effective population sizes ($N_e$) reduced far below census counts by sweepstakes reproduc-

tion [Hedgecock and Pudovkin, 2011]. This mismatch means the pathogen can track environmental changes orders of magnitude faster than the host can respond.

Compounding this asymmetry, the populations under strongest selection are also losing the genetic raw material needed for future adaptation. Additive genetic variance for resistance in central California drops 88% (from $V_A = 0.026$ to $V_A = 0.003$) as population bottlenecks eliminate rare alleles. Southern populations are being driven into an evolutionary corner: strong selection, but nothing left to select on.

The implication for conservation is clear. Captive breeding programs that screen for high-resistance broodstock [Hodin et al., 2021] and maintain genetic diversity through managed crosses are not merely helpful supplements to natural recovery—they may be the *only* path to re-establishing *Pycnopodia* in the southern half of its range, where natural selection alone leads to an evolutionary dead end.

### 6.6 Limitations

We highlight several important caveats:

- **Alaska underprediction.** As discussed, the model produces 2–7% Alaskan recovery against ∼50% field estimates. This is the model's main weakness and likely reflects missing local processes (fjord stratification, freshwater lensing) that suppress *Vibrio* activity in these environments.
- **Genetic architecture.** The 51-locus, three-trait framework is borrowed from GWAS data on *Pisaster ochraceus* [Schiebelhut et al., 2018]. The true genetic basis of SSWD resistance in *Pycnopodia* is unknown and may involve different numbers of loci, epistatic interactions, or trait architectures not captured by the additive model.
- **Climate projections.** The delta method applied to 11 CMIP6 reference sites cannot capture fine-scale oceanographic features—local upwelling intensification, mesoscale eddies, or freshwater-driven thermal anomalies—that may meaningfully alter SST trajectories at specific sites.
- **Simplified dispersal.** Larval and pathogen dispersal use distance-decay kernels without explicit ocean current data. Real connectivity patterns are shaped by coastal currents, seasonal reversals, and mesoscale features that our kernels cannot resolve.
- **Single pathogen.** We model *V. pectenicida* as the sole disease agent. The true SSWD etiology may involve co-infections, opportunistic secondary pathogens, or microbiome dysbiosis that amplifies the primary infection.
- **Resistance as a scalar trait.** Real disease resistance arises from complex immune networks with epistasis, dominance, and genotype-by-environment interactions. Our additive scalar approximation captures the broad dynamics of selection but not the full complexity of host–pathogen coevolution at the molecular level.

### 6.7 What comes next

Three immediate extensions of this work are planned:

**Reintroduction scenarios (F05–F08):** We will simulate captive breeding releases at depleted southern sites, testing different release sizes, frequencies, broodstock resistance levels, and

site selection strategies. These scenarios will directly inform the operational decisions facing the *Pycnopodia* recovery program.

**Sensitivity analysis:** A formal Morris screening followed by Sobol variance-based decomposition will identify which model parameters most strongly influence population outcomes. This will focus future calibration effort where it matters most and quantify the uncertainty envelope around our forecasts.

**Publication manuscript:** The present report will be condensed and submitted for peer review, providing the broader marine conservation community with a quantitative framework for *Pycnopodia* recovery planning.

Taken together, the forecast results paint a picture that is challenging but not hopeless. Natural recovery alone will not restore *Pycnopodia* to its former abundance anywhere in its range, and southern populations face functional extinction without intervention. But the model also identifies genuine bright spots—Oregon's thermal refuge, Alaska's cold-water reservoir, and the steady (if slow) accumulation of resistance alleles in surviving populations—that provide conservation managers with concrete targets for monitoring, protection, and eventual reintroduction.

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# A Region Codes

Table 4 maps the abbreviated region codes used throughout this report to their full geographic names.

Table 4: Region code glossary. Regions are listed north to south.

| Code | Full Name | Approx. Latitude |
|---|---|---|
| AK-PWS | Prince William Sound, Alaska | 60.5°N |
| AK-EG | Eastern Gulf of Alaska | 59.5°N |
| AK-OC | Alaska Outer Coast | 59.0°N |
| AK-FN | Fairweather North (SE Alaska) | 58.5°N |
| AK-WG | Western Gulf of Alaska | 57.5°N |
| AK-FS | Fairweather South (SE Alaska) | 57.0°N |
| AK-AL | Aleutian Islands | 53.0°N |
| BC-N | British Columbia North | 54.0°N |
| BC-C | British Columbia Central | 51.0°N |
| SS-N | Salish Sea North | 49.0°N |
| JDF | Juan de Fuca Strait | 48.3°N |
| SS-S | Salish Sea South | 48.0°N |
| WA-O | Washington Outer Coast | 47.0°N |
| OR | Oregon | 44.5°N |
| CA-N | California North | 40.5°N |
| CA-C | California Central | 36.5°N |
| CA-S | California South | 34.0°N |
| BJ | Baja California | 30.0°N |

# B Model Parameters

Table 5 lists all model parameters used in the W154 production configuration. Parameters marked with † were actively varied during the 154-round calibration process.

Table 5: Complete model parameters for the W154 production configuration.

| Parameter | Symbol | Value | Description |
|---|---|---|---|
| **Disease Transmission** | | | |
| Exposure coefficient | $a_{\text{exp}}$ | 0.75 | Efficiency of environmental pathogen translating to infection risk |
| Half-saturation† | $K_{\text{half}}$ | 800,000 | Environmental pathogen level for 50% of maximum dose–response |
| Pathogen activation | $T_{\text{vbnc}}$ | 12°C | Temperature above which dormant *Vibrio* become infectious |
| VBNC steepness | $k_{\text{vbnc}}$ | 2.0 | Sigmoid sharpness around $T_{\text{vbnc}}$ |
| Thermal floor | $T_{\text{vbnc,min}}$ | 9°C | Lowest temperature pathogen can evolve to activate at |
| **Environmental Pathogen Pool** | | | |
| Background input | $P_{\text{env,max}}$ | 2,000 | Maximum background *Vibrio* production (bact/mL/d) |

*Continued on next page*

| Parameter | Symbol | Value | Description |
|---|---|---|---|
| Accumulation† | $\alpha_{\text{env}}$ | 0.18 | Fraction of host shedding entering persistent pool |
| Decay rate | $\delta_{\text{env}}$ | 0.02 | Daily environmental pathogen degradation (2%/d, $t_{1/2} \approx 35$ d) |
| Pool floor | $P_{\text{floor}}$ | 500 | Minimum pathogen maintained at active sites |
| **Wavefront Spread** | | | |
| Origin sites | — | 5 | Channel Islands nodes where SSWD first appeared (2013) |
| Dose threshold | CDT | 1,000 | Cumulative pathogen dose to trigger local outbreak |
| Dispersal range | $D_P$ | 300 km | Characteristic waterborne pathogen dispersal distance |
| Max dispersal | $D_{P,\max}$ | 3,000 km | Maximum long-range pathogen transport |
| **Disease Progression** | | | |
| Pre-symptomatic shedding | $\sigma_1$ | 5.0 | Pathogen shed rate, $I_1$ stage |
| Symptomatic shedding | $\sigma_2$ | 50.0 | Pathogen shed rate, $I_2$ stage (10× higher) |
| Death shedding | $\sigma_D$ | 15.0 | Pathogen burst on host death |
| E→$I_1$ rate | $\mu_{EI_1}$ | 0.233/d | Exposed to pre-symptomatic (at 20°C) |
| $I_1$→$I_2$ rate | $\mu_{I_1 I_2}$ | 0.434/d | Pre-symptomatic to symptomatic |
| $I_2$→Death rate | $\mu_{I_2 D}$ | 0.563/d | Symptomatic to death |
| Recovery rate | $\rho_{\text{rec}}$ | 0.05 | Base daily clearance probability (× individual $c$) |
| **Pathogen Evolution** | | | |
| Thermal adapt rate | — | 0.001/d | Speed of thermal threshold drift |
| Virulence adapt rate | — | 0.001/d | Speed of virulence drift |
| Max virulence | $v_{\max}$ | 0.70 | Upper virulence bound |
| **Host Genetics** | | | |
| Resistance loci | $n_R$ | 17 | Loci controlling infection hazard reduction |
| Tolerance loci | $n_T$ | 17 | Loci controlling $I_2$ survival extension |
| Recovery loci | $n_C$ | 17 | Loci controlling pathogen clearance |
| Initial resistance | $\bar{r}_0$ | 0.15 | Starting hazard reduction (85% susceptible) |
| Initial tolerance | $\bar{t}_0$ | 0.10 | Starting tolerance |
| Initial recovery | $\bar{c}_0$ | 0.02 | Starting clearance scalar ($\times\rho_{\text{rec}} = 0.001$/d) |
| Tolerance ceiling | $\tau_{\max}$ | 0.85 | Maximum survival extension fraction |
| **Reproduction** | | | |
| Peak spawning day | — | Day 50 | Peak at reference latitude (48.5°N) |
| Latitude shift | — | 3 d/° | Spawning shifts later with latitude |
| SRS Pareto shape | $\alpha_{\text{SRS}}$ | 1.35 | Reproductive skew (lower = more sweepstakes-like) |
| Settler survival | $s_0$ | 0.001 | Larval survival to juvenile (0.1%) |
| Post-spawn immunosuppression | — | 28 d | Increased susceptibility after spawning |
| **Spatial Connectivity** | | | |
| Larval dispersal | $D_L$ | 400 km | Characteristic larval dispersal distance |

*Continued on next page*

| Parameter | Symbol | Value | Description |
|---|---|---|---|
| Connectivity exponent† | $n_{\text{conn}}$ | 0.3 | Enclosedness→flushing nonlinearity |
| Self-recruit (open)† | $\alpha_{\text{open}}$ | 0.02 | Minimum larval retention (exposed coast) |
| Self-recruit (fjord)† | $\alpha_{\text{fjord}}$ | 0.70 | Maximum larval retention (deep fjord) |
| **Population Biology** | | | |
| Carrying capacity | $K$ | 5,000 | Maximum population per site (uniform) |
| Max body size | $L_\infty$ | 1,000 mm | Asymptotic arm length (von Bertalanffy) |
| Growth rate | $k_g$ | 0.08/yr | Growth toward max size |
| Min reproductive size | $L_{\min}$ | 400 mm | Minimum size to spawn |
| Natural senescence | — | 50 yr | Age at increased natural mortality |
| **Simulation** | | | |
| Start year | — | 2012 | Simulation start |
| Duration | — | 38 yr | Total simulation years (2012–2050) |
| Disease onset | — | Year 1 | Wavefront begins in simulation year 1 (2013) |
| SST transition | — | 2025 | Switch from OISST observations to CMIP6 projections |

# C Calibration Targets

The model was calibrated against regional recovery fractions estimated from available survey data and expert assessment (Table 6). RMSE was computed on log-transformed fractions to weight relative errors equally across the four-order-of-magnitude range of targets.

Table 6: Regional recovery targets used for model calibration.

| **Region** | **Target (%)** | **Basis** |
|---|---:|---|
| AK-PWS (Prince William Sound) | 50 | Substantial survey-documented recovery |
| AK-FN (Fairweather North) | 50 | Cold-water remote populations |
| AK-FS (Fairweather South) | 20 | Partial recovery; warmer than PWS |
| BC-N (British Columbia North) | 20 | Moderate fjord-coast recovery |
| SS-S (Salish Sea South) | 5 | Limited recovery in semi-enclosed waters |
| JDF (Juan de Fuca Strait) | 2 | Near-total loss |
| OR (Oregon) | 0.25 | Severe decline, minimal recovery |
| CA-N (California North) | 0.1 | Near-extirpation |

The production model (W154) achieves a log-RMSE of 0.504 against these targets. The primary remaining discrepancy is in the Alaskan regions: the model predicts 2–7% recovery where survey data suggest up to 50%. This is discussed in Section 6.

# D SST Projection Methodology

Sea surface temperature (SST) for 2012–2025 was drawn directly from the NOAA Optimum Interpolation SST (OISST) v2.1 dataset, providing daily satellite-derived temperatures on a 0.25° grid. Each model site was assigned the nearest OISST grid cell.

For 2026–2050, we used the **delta method** to generate site-specific projections from CMIP6 multi-model ensemble outputs. The approach preserves each site's observed climatological baseline while applying projected warming trends:

$$T_{\text{site}}(y, m) = \bar{T}_{\text{site,obs}}(m) + \left[T_{\text{ref,proj}}(y, m) - \bar{T}_{\text{ref,base}}(m)\right] \tag{2}$$

where $\bar{T}_{\text{site,obs}}(m)$ is the site's observed monthly climatology (2012–2024 mean), $T_{\text{ref,proj}}(y, m)$ is the CMIP6 projection for the nearest reference site in year $y$ and month $m$, and $\bar{T}_{\text{ref,base}}(m)$ is that reference site's historical baseline (2012–2024 mean).

Eleven CMIP6 reference sites were selected to span the species' range, from the Aleutian Islands (53°N) to Baja California (30°N). Each of the 896 model sites was assigned to its nearest reference site by great-circle distance. Projections were generated under both SSP2-4.5 (moderate emissions, used in F01–F03) and SSP5-8.5 (high emissions, used in F04).

The delta method ensures that each site retains its unique seasonal SST cycle and inter-annual variability structure (from the observational record), with only the long-term warming trend adjusted according to the climate scenario. This approach avoids the biases that arise from using raw CMIP6 output directly at coastal locations, where global climate models often have significant resolution-related errors.

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